

Dissecting Jet-Tagger Through Mechanistic Interpretability

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Abstract

Mechanistic interpretability seeks to reverse engineer a trained neural network by identifying the minimal subset of internal components. We perform a mechanistic interpretability analysis of the Particle Transformer architecture, trained on the Top Quark Tagging reference dataset, with the goal of identifying the computational circuit responsible for jet classification and characterizing the physical content of its internal representations. Combining zero ablation, path patching with two complementary on-manifold corruption strategies and linear probing of the residual stream, we identify a sparse six-head circuit that recovers the great majority of the full model performance while admitting a clean source-relay-readout interpretation. In this circuit, a single early layer head serves as the primary causal source, a cluster of middle-layer heads acts as relays selectively attending to hard pairwise substructure and a single late-layer head reads out the aggregated signal. Linear probes show that the residual stream is preferentially aligned with the energy correlator basis over the N -subjettiness basis, with the advantage surviving residualization against jet mass. Within the energy correlator basis, the model preferentially encodes 2-prong substructure observables over the 3-prong observables that are the formally correct targets for top tagging, indicating that the network has implicitly factorized the 3-prong classification task into the more accessible subproblem of identifying the hadronic W -boson decay. A per-layer trained probe further reveals that the apparent single step commitment of the model to a classification decision in the first class attention block is in fact a basis rotation, with the discriminating signal already saturating in the particle attention stack. These results demonstrate that mechanistic interpretability methods developed for natural language models can be used for jet physics classifiers and indicate that gradient descent may rediscover physically meaningful aspects of jet tagging without supervision.

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1 Introduction

Jets [1] in High Energy Physics (HEP) are a collection of collimated sprays of stable particles, produced as the relic of hard scattered processes involving partons (viz. splitting and fragmentation of quarks and gluons, followed by hadronization), as well as hadronic decay of other Standard Model (SM) particles like top quark (t), heavy flavor quarks (b, c), electroweak bosons (W/Z) and the Higgs (H). The task of tracing down the primary truth origin of the particle cascade, which eventually forms a jet, is known as *Jet Tagging*. It is a foundational task in HEP, whose utility spans across most of the physics studies in a hadron collider environment, such as the CERN Large Hadron Collider (LHC), including a wide range of precision measurements and new-physics searches [2, 3]. Deep learning has transformed this task to new frontiers. Successive architectures, starting from CNNs using jet images [4] to graph neural networks [5, 6, 7] to the Particle Transformer [8], have produced classifiers whose performance substantially exceeds that of any single hand-crafted physics driven observable, such as N-subjettiness [9, 10].

The identification of highly boosted top quarks from their hadronic decay products is one of the central classification tasks in collider physics at the LHC. A boosted top quark decaying as $t \rightarrow Wb \rightarrow q\bar{q}b$ produces a single large radius jet whose constituents encode the kinematics of a three-body decay. The discrimination of such jets from background jets initiated by light quarks and gluons is a classification task that has been extensively studied with both expert designed jet-substructure observables, including N -subjettiness and energy correlation functions [11, 12]. Modern deep learning architectures, that operate directly on the particle four momenta, consistently outperform these expert designed taggers, established extensively on benchmark datasets [13].

The Particle Transformer (ParT) [8] is among the most widely used jet-tagging architectures currently available. It augments the standard Transformer [14] self-attention mechanism with a learned pairwise interaction bias derived from pairwise features of every particle pair, allowing each attention head to express a learned attention pattern that is informed by the physical structure of the jet. Despite the strong empirical performance of such models, the internal computation by which they reach their classification decision has remained opaque. Explaining these inner workings via analysis of attention maps, Shapley values and information flow is an active area of research in machine learning-based jet studies.

Mechanistic interpretability [15, 16, 17] is a research program that aims to reverse-engineer the computations performed by neural networks by identifying the minimal set of components, called a *circuit*, that is causally responsible for a given behavior, and by characterizing the algorithm those components implement. Originally developed in the context of language models, this program has produced a toolkit of activation patching, path patching, logit lens, and linear probing techniques that has not previously been applied systematically to jet physics

classifiers. The present work bridges this gap.

We train a small Particle Transformer (4 particle attention layers, 4 attention heads per layer, embedding dimension 128, approximately 1.3M trainable parameters) on a random subset of the Top Quark Tagging reference dataset [13, 18] and apply a complete mechanistic interpretability analysis. The contributions of this work are as follows.

1. We identify a six-head circuit that recovers 97.3% of full model AUC and is shown to be significantly more performing than randomly sampled six-head subsets at the 96-th percentile of the random baseline distribution.
2. We establish the causal structure of the circuit through path patching, distinguishing a single *source* head, three *relay* heads, a *secondary source* head, and a single *readout* head. The sign pattern of direct effects is consistent across two on-manifold corruption strategies, and we document a structural incompatibility between off-manifold (Gaussian) corruption and the standard recovery-score formulation that we expect to be relevant for any kinematically narrow physics dataset.
3. Linear probing of the residual stream against the family of classical jet-substructure observables reveals two distinct findings. First, the model preferentially encodes the energy correlator basis over the N -subjettiness basis, with the advantage persisting after residualization of $D_2^{(\beta=1)}$ and τ_{32} against jet mass. Second, within the energy correlator basis the model preferentially encodes 2-prong observables ($C_2^{(\beta=1)}$, $D_2^{(\beta=1)}$) over 3-prong observables ($C_3^{(\beta=1)}$, $N_3^{(\beta=1)}$), despite the latter being the canonical targets for a top tagger. It appears that the model has implicitly factorized the 3-prong top-tagging task into the more accessible 2-prong sub-task of identifying the hadronic W -boson decay.
4. Resolving the standard logit-lens analysis with per-layer trained logistic probes, we find that the apparent single-step commitment to a classification decision in the first class attention block is more like a *basis rotation*: linearly accessible class information is already present at $\text{AUC} \approx 0.97$ in the particle attention layers, and the class attention block aligns this signal with the basis of the final classification head.

The paper is structured as follows. Section 2 reviews the existing volume of literature with similar goals.

The relevant background on jet substructure and the Particle Transformer architecture are given in Appendix A.2. Section 3 reviews mechanistic interpretability. Section 4 describes the experimental setup. Section 5 presents the logit-lens analysis and its basis-rotation interpretation. Section 6 identifies and characterizes the circuit. Section 7 analyses the physical content of the residual stream, including the mass-residualized probe comparison. Section 8 examines the interaction features driving each head. Section 9 discusses the results in details and their implications. Finally we conclude in Section 10.

2 Existing Literature

Interpretability and explainability of machine learning models for collider physics have developed into an active subfield, motivated both by the desire to understand what these models compute and by the practical need to estimate systematic uncertainties on their outputs.

Existing work falls broadly into three categories: post-hoc attribution methods that assign importance scores to inputs of a trained model, interpretable-by-construction architectures that constrain the model’s function class, and recent attention pattern analyses of transformer-based jet taggers. To our knowledge, no prior work in collider physics has applied the full circuit level toolkit of mechanistic interpretability involving zero ablation, path patching, logit lens, and per-layer trained probes operating jointly on a single trained model, i.e. a jet classifier.

2.1 Post-hoc attribution and explainability methods

The earliest interpretability efforts for jet taggers were predominantly attribution based, identifying which input features or constituents most strongly influence the model’s prediction. Layer wise relevance propagation was applied to deep neural network jet taggers in combination with expert variables [19], confirming that learned models attribute importance to substructure observables consistent with theoretical expectations. A more detailed study of interpretability for DNN-based top taggers, combining layer wise relevance propagation, saliency maps, and gradient based attribution, was presented in Ref. [20]; this is the closest predecessor to the present work in topic, but it employs an entirely different methodological toolkit. Shapley value attribution was applied to boosted top tagging in Ref. [21], yielding per-input contributions to the classification score that satisfy the standard cooperative game axioms of additive feature attribution. Graph specific attribution methods, viz. GNNExplainer, GNNShap, and Grad-CAM, operating on the Lund jet plane were systematically compared recently in Ref. [22]. Beyond top tagging, attribution methods have been applied to particle-flow reconstruction [23], to multi-method explainability of muonic-pattern recognition [24] and to uncertainty quantification with evidential deep learning for jet identification, where epistemic uncertainty was used as a proxy for out-of-distribution detection [25]. The question of whether graph neural networks genuinely learn classical jet-substructure observables was addressed in Ref. [26] by training auxiliary regressors to predict observables from intermediate representations. This is a probing style methodology that is in spirit closest to the linear probing of Section 7 of the present work, but which was not coupled to a causal circuit analysis.

What distinguishes attribution methods from the present analysis is that they report *which inputs matter*, but not *how* the network combines them. A high relevance, saliency, or Shapley score for a particular constituent does not localize the computation in network space, does not establish causal information flow between internal components, and does not isolate a sparse sub-network that is itself sufficient for the behavior. The mechanistic interpretability program employed in the present work is designed precisely to address these complementary questions.

2.2 Interpretable-by-construction architectures

A second strand of work seeks interpretability not by analyzing a trained black box but by constraining the architecture so that its computation is transparent by design. Binary JUNIPR [27] phrases classification as a probabilistic model over the parton shower history, producing a discriminator whose internal log-likelihood ratios admit a direct physical reading at every node of the shower tree. Symbolic regression has been applied to extract closed form expressions for angular coefficients in Drell-Yan production [28], yielding equations that are interpretable by inspection at the cost of restricting the function class the model can express. The IRC-safe equivariant feature extractor of Ref. [29] enforces infrared and collinear safety as an architectural constraint, ensuring that learned representations respect the same

physical invariance properties as analytic substructure observables. Interpretable anomaly detection in jet substructure [30] likewise restricts the detector to a small expert designed feature space so that the learned decision boundary can be inspected directly. More recently, the Mixture-of-Experts graph transformer architecture of Ref. [31] introduces interpretability through gating. In this method, each expert specializes in a distinct kinematic regime, and the gating distribution serves as an interpretable summary of the learned partitioning of input space.

These approaches buy interpretability through architectural restriction. The present work occupies a complementary position: rather than constraining the architecture, we take a state-of-the-art unrestricted Particle Transformer and ask whether its internal computation can be reverse engineered after training. The two strategies may turn out to be complementary. Insights from circuit level analyses such as the one presented here can in principle inform the design of more interpretable physics aware architectures, and the source $\rightarrow$ relay $\rightarrow$ readout structure identified in Section 6 would be a natural template for an interpretable-by-construction successor.

2.3 Transformer-specific interpretability for jet tagging

The most direct neighbors of the present work are recent analyses of transformer-based jet taggers. The physical content of ML quark-vs-gluon discriminators has been characterized in Ref. [32] through correlation studies and feature importance analyses, establishing which observables trained taggers exploit without localizing the computation to specific architectural components. The interaction aware Transformer of Ref. [33] introduces architectural modifications informed by physics priors and inspects the resulting attention patterns, providing input-attribution-style insight without causal interventions on the trained model. Closer still are Refs. [34, 35, 36]: Ref. [34] interprets transformers for jet tagging through attention visualization and clustering of head behaviors, identifying head-specialization patterns; the follow-up of Ref. [35] studies the empirical sparsity of attention in the Particle Transformer specifically, characterizing which particle pairs receive non trivial attention weight as a function of layer and head; and Ref. [36] applies probing and feature attribution methods to a transformer trained on physics data and asks what physical content the trained model has acquired.

These transformer specific analyses share with the present work a focus on the internal structure of attention based jet classifiers, but they stop short of the causal toolkit deployed here. None performs path patching to establish causal information flow between heads, none distinguishes direct from indirect effects through systematic intervention, and none confronts the basis rotation ambiguity in the standard logit lens that the per layer trained probes of Section 5.2 resolve. The present work extends these efforts by combining attention pattern analysis, consistent in spirit with Refs. [34, 35], with the causal intervention methodology of mechanistic interpretability, and by closing the loop with a quantitative comparison between the encoded representation and the classical energy-correlator basis of jet substructure.

3 Mechanistic interpretability

Mechanistic interpretability [15] seeks to decompose a neural network’s computation into human interpretable components. The central objects of study are *circuits*, a subset of the computational graph (here attention heads and their connections) that are causally responsible

for a specified behavior. Several intervention tools are used in the present work.

Zero ablation. In this method, the contribution of a component is removed by setting its output to zero either by overwriting its activation or in the case of head scaling architectures, by setting the head’s scaling weight to zero. The resulting drop in some scalar performance measure quantifies how structurally important the component is.

Path patching. The prescription of this method is as following : the model is run on a *clean* input and a *corrupted* input. The output of a chosen component is recorded on the clean run and substituted into the corresponding position of the corrupted run, with all other components processing the corrupted input. Let $LD = \log p_{\text{top}} - \log p_{\text{QCD}}$ denote the logit difference (with p_{top} and p_{QCD} being the output probability score of the tagger for the binary classification between two classes of *top* and *QCD*), which serves as the scalar measure of the model’s confidence in the top hypothesis. The *direct effect* of a head (l, h) is the recovery score

$$DE(l, h) = \frac{LD_{\text{patched}}(l, h) - LD_{\text{corrupt}}}{LD_{\text{clean}} - LD_{\text{corrupt}}}. \quad (1)$$

where $LD_{\text{patched}}(l, h)$ denotes the logit difference of the model run on the corrupted input with the activation of head (l, h) replaced by its value from the clean run, and $LD_{\text{clean}}, LD_{\text{corrupt}}$ denote the logit differences on the clean and corrupted runs respectively without any patching. A direct effect of +1 corresponds to the patched run reproducing the clean logit difference exactly; values larger than +1 correspond to super-recovery, in which the patched activation drives the logit difference further past its clean-run value than the clean run itself. On the dataset studied here the within-batch replacement corruption yields $\overline{LD}_{\text{corrupt}} > \overline{LD}_{\text{clean}}$ (corrupted top jets are mildly more top-like, since the replacing particles are themselves drawn from a 50%-top sample; see Section 6.2), so the denominator of Eq. (1) is negative. Within this sign convention, a negative direct effect indicates that transplanting the head’s clean-run activation drives the patched run’s logit difference *above* the unpatched corrupted baseline, i.e., that the head’s clean-run output is incompatible with the rest of the corrupted run.

The *path effect* between a source head (l_s, h_s) and a target head (l_t, h_t) with $l_s < l_t$ is the L^2 norm of the difference between the target head’s output under (i) the corrupted run and (ii) the corrupted run with only the source head’s output replaced by its clean-run value. This isolates the contribution that flows from the source head to the target head along the residual stream and through the intervening architecture.

Logit lens. The logit lens [37] projects an intermediate representation through the final layer normalization and the trained classification head, producing a pseudo prediction at each layer. The area under the curve (AUC) of the resulting pseudo prediction quantifies the amount of class information that is linearly accessible to the trained head at that depth. A limitation of the logit lens is that it assumes the intermediate representation is expressed in the same basis as that on which the head was trained; we revisit this point in Section 5.

Linear probing. Linear probing [38] method fits a small linear model (Ridge regression [39] for continuous targets, logistic regression for binary targets) to predict a target observable from the intermediate representation. The held out coefficient of determination R^2 (or AUC, for binary targets) measures the linear accessibility of that observable in the representation.

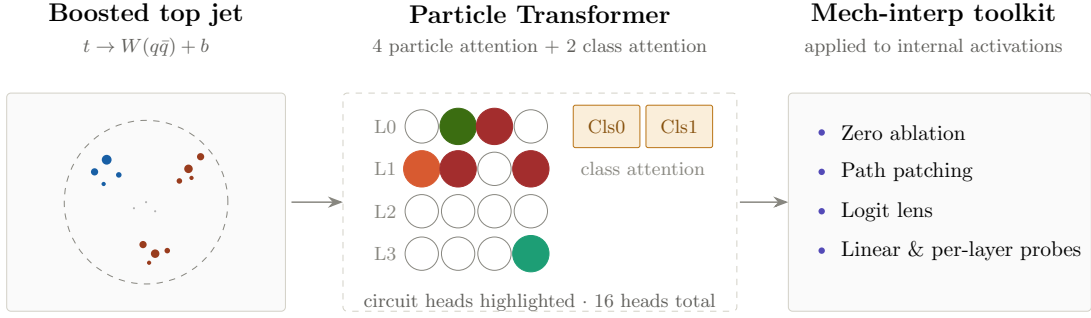


Figure 1: Schematic overview of the analysis pipeline. A boosted top jet ($t \rightarrow W(q\bar{q}) + b$, left) is processed by a Particle Transformer with four particle attention layers and two class attention layers, for a total of sixteen attention heads (center); the six heads identified in this work as constituting the classification circuit are filled, with the remaining ten heads shown as empty circles. The mechanistic interpretability toolkit applied to the model’s internal activations (right) comprises zero ablation, path patching, the logit lens, and linear and per-layer probes of the residual stream. The detailed direct-effect values and information-flow paths within the circuit are presented in Figure 7.

As with the logit lens, the result is a lower bound on the true information content, because non linearly encoded information is not detected.

The mechanistic interpretability of the ParT is shown in the Figure 1.

4 Experimental Setup

4.1 Dataset

We use the Top Quark Tagging reference dataset of Refs. [13, 18], which consists of simulated jets from pp collisions at $\sqrt{s} = 14$ TeV. Signal jets are initiated by hadronic top decays, $t \rightarrow Wb \rightarrow q\bar{q}b$; background jets are initiated by light quarks and gluons. Events are simulated with PYTHIA 8 [40] and detector effects are modeled by a fast parametric simulation. Both classes are selected with jet transverse momentum $p_T \in [550, 650]$ GeV and pseudo-rapidity $|\eta| < 2$. Jets are reconstructed with the anti- k_T algorithm [41] using FastJet [42] with radius parameter $R = 0.8$. Each jet is represented by up to 200 constituent particles ordered by decreasing p_T , with zero-padding for jets containing fewer constituents.

We use 200,000 jets for training, 50,000 for validation, and 50,000 for testing, with a 50% signal fraction in all splits. All mechanistic interpretability analyses use the test set unless otherwise stated. The full dataset is not used primarily due to computationally intensive nature of the study.

4.2 Model architecture

We train a ParT with 4 particle attention layers, 2 class attention layers, 4 attention heads per layer, embedding dimension $d = 128$, feed-forward expansion ratio 4, and pairwise interaction embedding dimensions $[64, 64, 64, 4]$ (the final dimension matching the number of attention

heads). The total number of trainable parameters is approximately 1.3M. The necessary details of the ParT architecture can be found in Appendix A.2.

The per-particle input features for each constituent are: relative pseudo-rapidity $\Delta\eta$, relative azimuthal angle $\Delta\phi$, $\log p_T$, $\log E$, $\log(p_T/p_{T,\text{jet}})$, $\log(E/E_{\text{jet}})$, and ΔR from the jet axis. The pairwise interaction MLP receives the four pairwise features in Eqn. 14 computed from the constituent four-momenta.

We use two related indexing conventions throughout this paper. Attention heads are labelled $L\ell Hh$, where $\ell \in \{0, 1, 2, 3\}$ indexes the four particle attention layers and $h \in \{0, 1, 2, 3\}$ indexes the four heads within each layer. Residual stream depths are labelled Ld , where $d \in \{0, 1, 2, 3, 4\}$: $L0$ denotes the initial particle embedding prior to any attention layer, and Ld for $d \geq 1$ denotes the residual stream at the output of the d -th particle attention layer. The output of the layer containing heads $L\ell Hh$ therefore appears in the residual stream at depth $L(\ell + 1)$; for example, the primary source head $L0H1$ resides in the first particle attention layer, and its output is read out from the residual stream at depth $L1$.

4.3 Training

The model is trained with the AdamW optimizer [43] with learning rate 10^{-3} , $\beta_1 = 0.95$, $\beta_2 = 0.999$, $\epsilon = 10^{-5}$, and weight decay set to zero. A linear schedule decays the learning rate from 10^{-3} to 10^{-5} over the final 30% of training steps after a flat warmup. Training proceeds for 30 epochs with batch size 256, mixed-precision (FP16) computation, and gradient clipping at L^2 norm 1.0. The best checkpoint is selected by validation AUC.

We train five independent models with random seeds 0 through 4 to assess reproducibility. The test AUCs across the five seeds are $\{0.9796, 0.9800, 0.9796, 0.9785, 0.9793\}$, giving a mean and standard deviation of 0.9794 ± 0.0005 . This is below the AUC of the original full scale ParT on the same dataset subset (≈ 0.985), as expected for the much smaller model used here, but already in the regime where transformer-class architectures outperform classical jet substructure observables. All mechanistic interpretability analyses use the seed-0 model unless otherwise stated; multi seed results are reported where relevant.

4.4 Intervention methods

In this subsection we discuss in details the several intervention methods that have been used in the present study.

Zero ablation. The Particle Transformer used here implements NormFormer-style head scaling [44], where each attention head’s output is multiplied by a learned scalar $c_{l,h}$ before being summed into the residual stream. We ablate head (l, h) by setting $c_{l,h} = 0$ during inference, which removes the head’s contribution while leaving all other parameters unchanged. The importance is measured as the drop in the mean logit difference over top jets,

$$I_{\text{zero}}(l, h) = \overline{\text{LD}}_{\text{baseline}} - \overline{\text{LD}}_{\text{ablated}}(l, h), \quad (2)$$

averaged over 5,000 top jets and across all five training seeds.

Path patching: corruption strategy. Path patching requires a corrupted input that destroys the discriminating substructure while remaining within the kinematic regime on which the model was trained. The primary strategy used in this work is *within batch particle*

replacement: each top jet’s particles are replaced by those of another jet in the same batch (offset by half the batch size), preserving the per-particle feature distribution but breaking the jet level kinematic correlations. Direct effects are computed over 2,000 test jets; we verify that the denominator $LD_{\text{clean}} - LD_{\text{corrupt}}$ is non-zero and consistent in sign across the test sample.

To assess the robustness of the direct effect sign pattern, we additionally run path patching with a *whole-jet permutation* corruption that permutes intact jets across the batch (without mixing constituents). This is a milder corruption that preserves intra-jet correlations but breaks the top-vs-QCD label assignment.

A third strategy, *Gaussian noise corruption*, was attempted in two implementations, both of which proved to be structurally incompatible with the standard recovery-score formulation in Eq. (1); the underlying reason and its consequences for physics domain interpretability are discussed in Section 6.6.

Circuit masking. Finally in this method, for the partial minimality test and the random-baseline analysis, all heads *outside* a specified circuit are zero-ablated and the test AUC is evaluated on the full 50,000-jet test set. Bootstrap 95% confidence intervals on circuit AUCs are computed with 1,000 resamples.

5 When Does the Model Commit to Its Prediction?

5.1 Logit lens trajectory

The logit lens projects the residual stream at each depth through the trained final classification head, producing a pseudo-prediction whose AUC quantifies how much class-discriminating information is linearly accessible to the trained head at that layer. Figure 2 shows the logit-lens AUC at each depth in the network: the initial particle embedding (L0), the four particle attention layers (L1–L4), and the two class attention blocks (Cls0, Cls1).

The trajectory through the particle attention layers is highly non-monotone: $0.575 \rightarrow 0.366 \rightarrow 0.541 \rightarrow 0.801 \rightarrow 0.111$. The AUC at L4, the output of the last particle attention layer, is 0.111, substantially below the random baseline of 0.5. This is followed by a jump of $+0.862$ at the first class attention block (Cls0), with the second class attention block (Cls1) adding only $+0.006$.

Figure 3 shows the corresponding mean logit difference evaluated separately for top and QCD jets. The two classes have similar and near-zero logit differences throughout the particle attention layers and diverge sharply at Cls0, with top jets reaching a mean logit difference of approximately $+3.7$ and QCD jets dropping to approximately -2.5 .

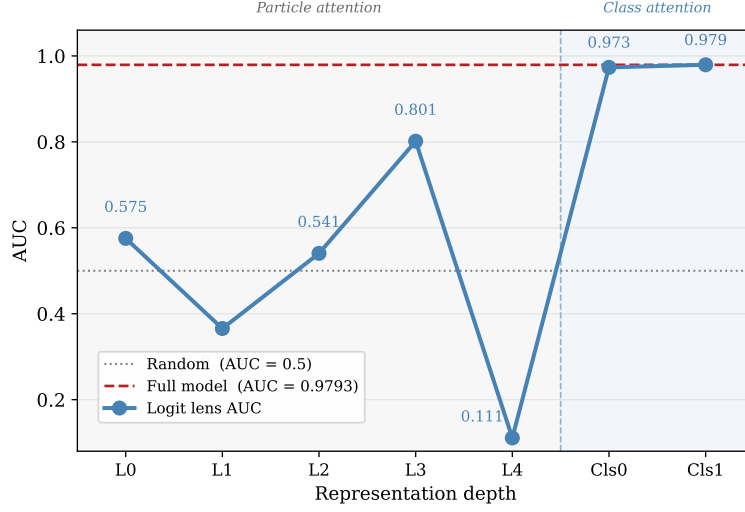


Figure 2: Logit-lens AUC as a function of representation depth. At each depth, the mean-pooled particle residual stream is projected through the trained final layer normalization and classification head to produce a pseudo-prediction. The dashed red line shows the full model test AUC of 0.9793. The shaded region marks the class attention blocks. The trajectory through the particle attention layers is non-monotone and ends at $\text{AUC} = 0.111$ at L4, well below the random baseline; a single-step jump to 0.973 occurs at the first class attention block (Cls0).

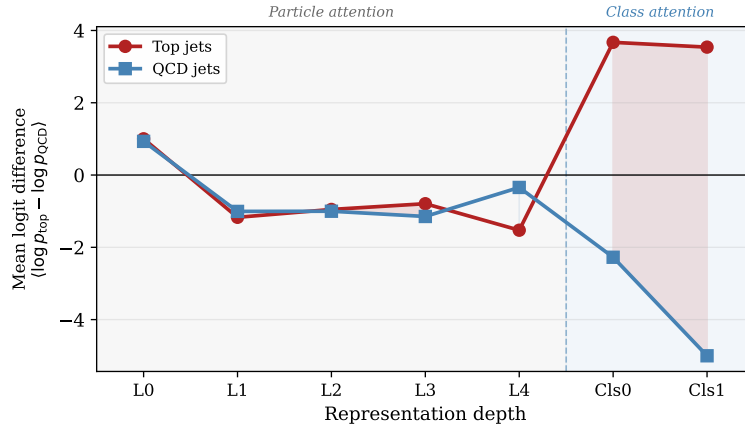


Figure 3: Mean logit difference $\langle \log p_{\text{top}} - \log p_{\text{QCD}} \rangle$ as a function of representation depth, evaluated separately for top jets (red) and QCD jets (blue) over 10,000 test jets. Through all particle attention layers the two classes have similar and near-zero logit differences. A sharp class separation emerges only at the first class attention block (Cls0).

Read at face value, these observations suggest that no class-discriminating signal is present in the particle attention layers and that the entire classification decision is computed within Cls0. This conclusion would, however, be premature: the logit lens projects intermediate representations through a layer normalization and classification head that were trained specifically on the output of the class attention stack, and a low logit-lens AUC at an intermediate layer

can equally well indicate that the information is present but in a basis incompatible with the trained head. We resolve this ambiguity in the next subsection.

5.2 The class attention block as a basis rotation

To distinguish absent class information from misaligned class information, we train a per-layer logistic regression probe directly on each layer’s mean-pooled representation, with the same input-output pairs used for the logit lens. The probe is a single linear layer with its own implicit normalization (a standard scaler) and is fitted to maximize binary classification accuracy on a 7,000-jet training subset of the test set, with AUC evaluated on the remaining 3,000 jets.

Table 1 compares the logit-lens AUC and the per-layer probe AUC at each depth, and Figure 4 shows the comparison graphically.

Table 1: Logit-lens AUC and per-layer trained logistic probe AUC at each representation depth, evaluated on the same 10,000 test jets. The logit lens uses the trained final classification head; the per-layer probe is trained from scratch on each layer. A large discrepancy at a particle attention layer indicates that class-discriminating information is present in the representation but in a basis to which the trained head is not aligned.

Depth	Logit lens	Per-layer probe	Difference
L0	0.575	0.918	+0.343
L1	0.366	0.971	+0.605
L2	0.541	0.974	+0.434
L3	0.801	0.976	+0.174
L4	0.111	0.977	+0.866
Cls0	0.973	0.976	+0.003
Cls1	0.979	0.978	−0.002

The per-layer probe AUC reaches 0.971 already at L1 and remains in the range 0.971-0.977 throughout the particle attention layers. At L4, where the logit lens reports $\text{AUC} = 0.111$, the trained per-layer probe achieves $\text{AUC} = 0.977$, only marginally lower than the full model’s 0.9793. The two metrics agree to within 0.003 at Cls0 and Cls1.

The mechanistic interpretation that emerges is qualitatively different from the standard logit-lens reading. The class-discriminating signal saturates within the first particle attention layer; the residual stream of L1-L4 contains essentially all the linearly accessible class information that the network ever produces. The role of the first class attention block is then a *basis rotation*: a single global attention operation that reorients the latent class signal from the particle attention basis (orthogonal to, and partially anti-aligned with, the trained head) into the basis on which the head was trained. The dramatic +0.862 logit-lens jump at Cls0 measures the magnitude of this rotation, not a discontinuous gain in information content.

This refinement is consistent with the linear probe analysis of Section 7, which finds that the binary is_top target is encoded with similar R^2 at L4 and at the class attention blocks. It also clarifies the role of the second class attention block Cls1, which adds essentially no logit-lens AUC (+0.006) over Cls0: once the basis has been aligned, no further computation is necessary.

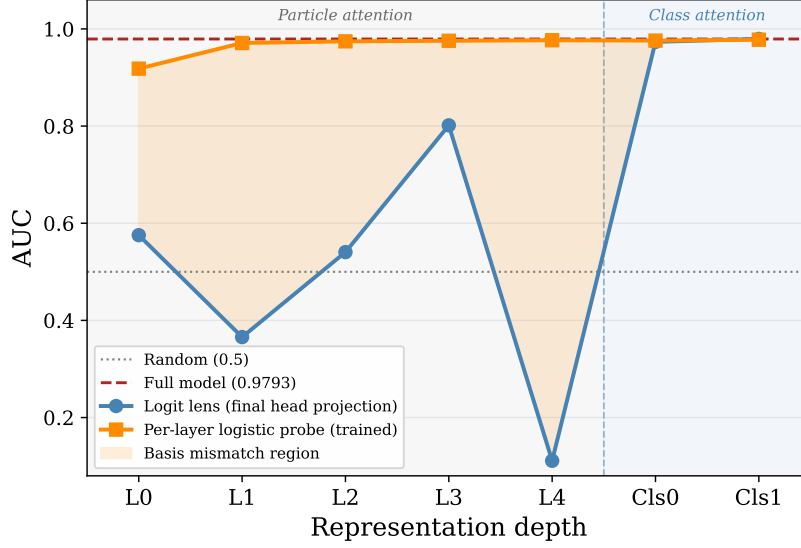


Figure 4: Comparison of logit-lens AUC (blue circles) and per-layer trained logistic probe AUC (orange squares) as a function of representation depth. The shaded region between the two curves quantifies the basis mismatch between the layer’s representation and the trained classification head. Class-discriminating information is essentially complete by L1 (AUC = 0.971) and remains so through L4 (AUC = 0.977), even where the logit lens reports values below random (AUC = 0.111 at L4). The two metrics agree at the class attention blocks (Cls0, Cls1).

6 Circuit Identification

Having established that the class discriminating signal is constructed within the particle attention stack rather than the class attention block, we now ask which of the model’s sixteen attention heads are responsible for this computation and how information flows between them. We address this in three stages: zero ablation to identify a candidate set of important heads, path patching to establish the causal structure connecting them, and minimality and random-baseline tests to confirm that the resulting circuit is both sufficient and non trivial.

6.1 Head importance from zero ablation

We begin by ranking the sixteen attention heads by their structural importance, measured as the drop in mean logit difference when each head’s output is set to zero. Figure 5 reports the zero-ablation importance $I_{\text{zero}}(l, h)$ for all 16 attention heads, averaged over the five training seeds.

Two heads stand out clearly: L0H1 with $I_{\text{zero}} = 1.491 \pm 0.051$ and L1H3 with $I_{\text{zero}} = 1.469 \pm 0.054$. The standard deviations across seeds lie in the range 0.050–0.056, and the ranking of the heads is consistent across all five seeds. Twelve of the sixteen heads are significant at the 2σ level; the four non-significant heads are L2H0 (-0.113 ± 0.054), L2H1, L2H2, and L3H1, all in Layer 2 or in a single Layer 3 head. The top six heads by importance are L0H1, L1H3, L0H2, L1H0, L1H1, and L3H3. These constitute the candidate circuit studied in the remainder of this section.

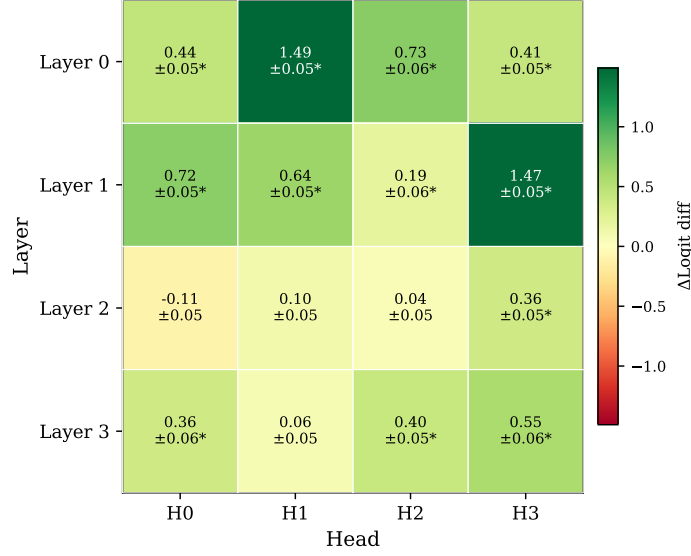


Figure 5: Head importance measured by zero ablation, averaged over five independent training seeds. Each cell shows the mean and standard deviation of $I_{\text{zero}}(l, h)$, with asterisks marking heads significant at the 2σ level. The model AUC is consistent across seeds (0.9794 ± 0.0005). The two most important heads are L0H1 (1.49 ± 0.05) and L1H3 (1.47 ± 0.05).

The cross-seed stability of the importance ranking is a non-trivial feature: a circuit identified in a single trained model could in principle be an artifact of that particular random initialization. Here, the identity of the top-six heads varies only in ordering across seeds and the rankings are stable to within their statistical uncertainties. This provides a first piece of evidence that the circuit reflects a property of the training objective and dataset rather than the random initialization.

6.2 Causal structure from path patching

Zero ablation measures the structural importance of each head but does not distinguish between heads that carry discriminating information independently and heads that are important only in the context of other heads. Path patching provides this distinction by causally intervening on individual head outputs.

We perform path patching on the seed-0 model with the within-batch replacement corruption, computing direct effects over 2,000 test jets. The clean and corrupted mean logit differences are $\overline{\text{LD}}_{\text{clean}} = +0.021$ and $\overline{\text{LD}}_{\text{corrupt}} = +0.089$, giving the denominator $\overline{\text{LD}}_{\text{clean}} - \overline{\text{LD}}_{\text{corrupt}} = -0.068$. The negative sign indicates that on the corrupted runs the model becomes mildly more confident in the top hypothesis, a kinematic consequence of replacing a top jet’s particles with those of another jet that is itself drawn from a 50%-top dataset.

Table 2 lists the direct effect of each circuit head, and Figure 6 shows the full 4×4 direct-effect matrix.

The direct effect pattern reveals a three-tier structure that we classify as follows. L0H1 is a *primary source*: a single head whose clean-run output, transplanted alone, is sufficient to recover most of the discriminating signal ($\text{DE} = +4.25$, i.e., super-recovery). L3H3 is a *readout*

Table 2: Direct effects of the six circuit heads from path patching, averaged over 2,000 test jets. Positive values indicate that transplanting the head’s clean run output into a corrupted run recovers the clean logit difference; negative values indicate that the transplanted activation is incompatible with the rest of the corrupted run. The role assignment is read off the sign and magnitude pattern in combination with the path effect graph (Section 6.3) and the representational similarity analysis (Section 6.7).

Head	Direct effect	Role
L0H1	+4.25	Primary source
L0H2	−1.92	Secondary source
L1H0	−1.13	Relay
L1H1	−2.95	Relay
L1H3	−2.48	Relay
L3H3	+0.71	Readout

head: its modest positive direct effect (+0.71) is consistent with an aggregation operation that benefits from, but does not produce, the underlying class signal. L0H2 (DE = −1.92) and the three Layer 1 heads L1H0, L1H1, L1H3 (DE = −1.13, −2.95, −2.48) all have negative direct effects, indicating that their clean-run activations are conditional on the upstream signal from L0H1 and that transplanting any one of them in isolation introduces a representational inconsistency that degrades performance. The cosine similarity analysis of Section 6.7 groups L0H2 with L0H1 (cosine similarity 0.92), placing L0H2 in a *secondary source* role rather than a relay role; this taxonomy is used consistently throughout the remainder of the paper.

6.3 Path effects with bootstrap confidence intervals

Direct effects identify which heads carry causal weight but do not specify how information flows between them. This is given by path effects, defined as the L^2 representational delta between corrupted and source-patched target outputs.

Table 3 reports the eleven forward path effects in the candidate circuit, with 95% bootstrap confidence intervals computed from 500 bootstrap resamples over 2,000 test jets.

The eleven path effects fall into three non-overlapping groups. L0H1 sends path effects of 1.19–1.45 to its four downstream targets (the three Layer 1 relay heads and the L3H3 readout). L0H2 sends weaker path effects of 0.85–0.90 to the same targets. The Layer 1 relay heads in turn send path effects of 0.68–0.75 to the L3H3 readout. The hierarchy primary-source $\succ$ secondary-source $\succ$ relay-to-readout is statistically resolved: the smallest L0H1-sourced path effect (1.19) is more than 25 standard deviations above the largest L0H2-sourced one (0.90).

The structure suggested by these path effects is summarised in Figure 7.

6.4 Minimality and sufficiency

A true test of minimality for the circuit would require an analysis using methods like ACDC. We do not do such a minimality test in this work. We test the sufficiency of the six-head circuit by adding heads one at a time in order of decreasing recovery score and evaluating the masked-circuit AUC on the full test set, with bootstrap 95% confidence intervals. Table 4 reports the results.

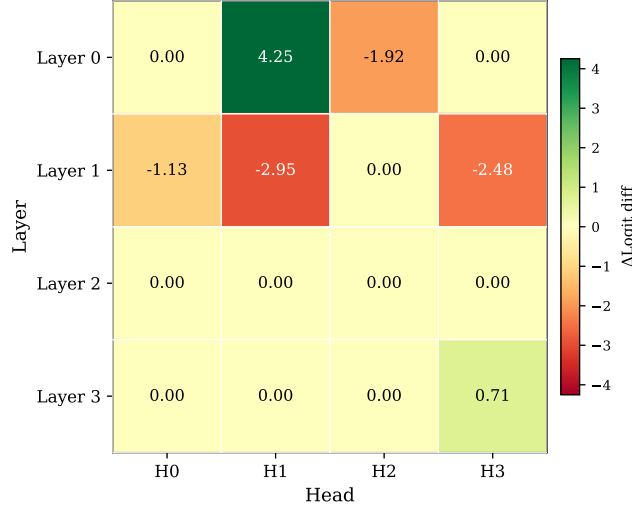


Figure 6: Direct effect (path-patching recovery score) for all 16 heads. Heads not in the candidate circuit have direct effect set to zero by convention since they were not patched. The dominance of L0H1 (+4.25), the negative effects of the four Layer 0 and Layer 1 relay-type heads, and the small positive direct effect of L3H3 (+0.71) are visible.

The trajectory has two notable nonlinearities. L0H1 alone achieves 88.6% of the full model AUC, consistent with its dominant direct effect. The most significant non monotonic feature occurs at step 5, when L1H3 is added: the AUC jumps from 0.909 to 0.947, an increment of +3.8 percentage points. This is substantially larger than any of the preceding steps, despite L1H3 having the second-most-negative direct effect (-2.48). The interpretation is that L1H3 functions effectively only when the upstream context (L0H1, L3H3, L1H0, L0H2) is already in the circuit; this pattern of conditional sufficiency is the defining signature of a relay node and distinguishes the present circuit from a collection of independent components.

6.5 Random-baseline comparison

To assess whether the identified circuit is significantly more performant than an arbitrary six-head subset, we compare its AUC against a baseline of 200 randomly sampled six-head subsets. The remaining heads in each random subset are masked, and the test AUC is computed on the full test set with all other settings identical.

The random six-head subsets achieve a mean AUC of 0.869 with standard deviation 0.125, compared to the identified circuit’s 0.953 (Figure 9). The identified circuit lies at the 96-th percentile of the random distribution: of the 200 random subsets, only 8 achieve an AUC at or above 0.953. We refrain from quoting a parametric z -test p -value here because the broad spread of the random distribution makes the empirical percentile a more directly interpretable summary; what matters is that the identified circuit is significantly more performant than typical sparse subsets while not being uniquely so. The substantial spread of the random distribution, with several random subsets achieving AUC above 0.93 but others falling below 0.65, is informative in its own right and indicates that the model has learned a somewhat redundant representation in which many sparse subsets retain substantial task-relevant information while many others lose it entirely. The merit of the identified circuit is the combination

Table 3: Forward path effects (representational deltas) within the candidate circuit, with 95% bootstrap confidence intervals over 500 bootstrap resamples on 2,000 test jets. The three natural groupings (primary source $\rightarrow$ Layer 1, secondary source $\rightarrow$ Layer 1, Layer 1 $\rightarrow$ readout) are pairwise non-overlapping, establishing a quantitative ordering of the information flow.

Path	Mean Δ	95% CI lower	95% CI upper
L0H1 $\rightarrow$ L1H1	1.454	1.417	1.494
L0H1 $\rightarrow$ L1H3	1.404	1.360	1.443
L0H1 $\rightarrow$ L3H3	1.351	1.318	1.383
L0H1 $\rightarrow$ L1H0	1.189	1.160	1.217
L0H2 $\rightarrow$ L1H0	0.900	0.881	0.919
L0H2 $\rightarrow$ L1H1	0.858	0.839	0.876
L0H2 $\rightarrow$ L3H3	0.855	0.838	0.872
L0H2 $\rightarrow$ L1H3	0.850	0.833	0.867
L1H3 $\rightarrow$ L3H3	0.754	0.735	0.771
L1H0 $\rightarrow$ L3H3	0.752	0.736	0.769
L1H1 $\rightarrow$ L3H3	0.683	0.668	0.698

of its strong AUC *and* its clean mechanistic interpretation as a source-relay-readout structure (an interpretation that random subsets do not, in general, admit).

A complete (though much more computationally intensive) enumeration of all $\binom{16}{6} = 8008$ six-head subsets would replace the random-baseline percentile with an exact rank.

6.6 Robustness of the direct-effect sign pattern under alternative corruption strategies

Path-patching results depend on the choice of corruption strategy (Section 4.4). To test whether the sign pattern of the direct effects is robust, we re-run the analysis on 2,000 test jets with the whole-jet permutation corruption. Table 5 reports the result.

The sign pattern is fully consistent: L0H1 and L3H3 have positive direct effects under both strategies; L0H2 and the three relay heads have negative direct effects under both. Magnitudes are uniformly smaller under permutation than under replacement, which is consistent with the smaller absolute value of the denominator $\text{LD}_{\text{clean}} - \text{LD}_{\text{corrupt}}$ on this milder corruption.

Gaussian noise corruption. We additionally explored Gaussian noise corruption in two implementations: additive noise on the four-momentum components (p_x, p_y, p_z, E) , and additive noise in the kinematic representation $(\log p_T, \eta, \phi)$ followed by reconstruction of physically valid four-vectors satisfying $E = |\mathbf{p}|$. At every noise scale that we examined (from 0.05 to 2.0 times the empirical kinematic standard deviation), Gaussian corruption drives the model into a strongly QCD-confident regime, with $\text{LD}_{\text{corrupt}} \ll 0$. This inverts the sign of the denominator $\text{LD}_{\text{clean}} - \text{LD}_{\text{corrupt}}$, which is negative under the on-manifold strategies (Section 6.2) and would be positive under Gaussian corruption. The recovery scores from the two regimes are therefore not on a common footing, and Gaussian-corruption results cannot be combined with the on-manifold ones without redefining the recovery score. We attribute this to the kinematically narrow training distribution of the dataset ($p_T \in [550, 650]$ GeV) and to the

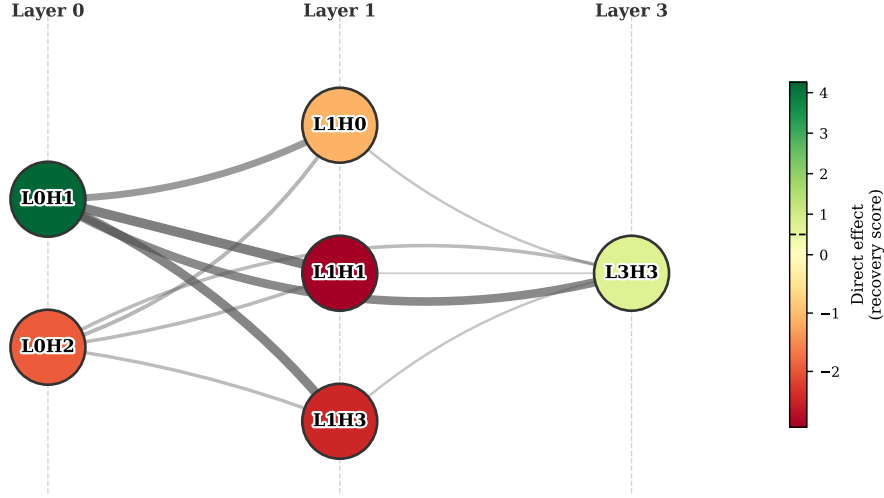


Figure 7: The identified circuit. Nodes represent the six circuit heads; node color encodes the direct effect on a diverging scale (green: positive; red: negative); edge thickness encodes the path-effect magnitude. The circuit has a source-relay-readout structure: L0H1 (primary source, $DE = +4.25$) sends the strongest path effects to three Layer 1 relay heads, which collectively forward information to L3H3 (readout, $DE = +0.71$). L0H2 (secondary source, $DE = -1.92$) sends weaker but still substantial path effects to the same relay heads. No Layer 2 head appears in the circuit.

absence of data augmentation: off-manifold inputs at any noise scale are sufficiently far from the training distribution that the model collapses to a fixed, strongly negative logit difference. We suspect this to be a generic feature of any physics domain interpretability analysis on a tightly distributed dataset and we recommend on-manifold corruption strategies as the default choice.

6.7 Representational similarity

The cosine similarity between the mean jet level output representations of each circuit head pair, computed over 3,000 test jets, is shown in Table 6.

Table 4: Partial minimality and sufficiency test. Heads are added to the circuit in order of decreasing recovery score. At each step, all heads outside the current circuit are masked and the test AUC is evaluated on the full 50,000-jet test set. Bootstrap 95% confidence intervals are computed with 1,000 resamples. The six-head circuit recovers 97.3% of the full model AUC of 0.9793.

Step	Head	DE	AUC	95% CI	% of full model
1	L0H1	+4.25	0.8676	[0.8641, 0.8710]	88.6%
2	L3H3	+0.71	0.8921	[0.8888, 0.8953]	91.1%
3	L1H0	-1.13	0.9054	[0.9026, 0.9079]	92.5%
4	L0H2	-1.92	0.9089	[0.9062, 0.9114]	92.8%
5	L1H3	-2.48	0.9469	[0.9450, 0.9488]	96.7%
6	L1H1	-2.95	0.9527	[0.9510, 0.9544]	97.3%

Table 5: Direct effects of the six circuit heads under two on-manifold corruption strategies: within batch particle replacement (the primary strategy used throughout this work) and whole jet permutation. The qualitative sign pattern, viz., positive for the primary source L0H1 and the readout L3H3, negative for the secondary source L0H2 and all three relay heads, is fully consistent across the two strategies. Magnitudes are smaller under jet permutation than under replacement, which we attribute to the smaller absolute denominator $LD_{\text{clean}} - LD_{\text{corrupt}}$ achieved by the milder corruption.

Head	Role	Replacement	Jet permutation
L0H1	Primary source	+4.25	+3.78
L0H2	Secondary source	-1.92	-1.72
L1H0	Relay	-1.13	-0.67
L1H1	Relay	-2.95	-2.51
L1H3	Relay	-2.48	-2.17
L3H3	Readout	+0.71	+0.66

Table 6: Cosine similarity between the mean jet-level output representations of each circuit head pair, computed over 3,000 test jets. Two clusters emerge: a Layer 0 cluster {L0H1, L0H2} with cosine similarity 0.92, and a Layer 1 cluster {L1H0, L1H1, L1H3} with pairwise similarities in [0.81, 0.95]. The L3H3 readout has low similarity to all other heads.

	L0H1	L0H2	L1H0	L1H1	L1H3	L3H3
L0H1	1.000	0.922	0.027	0.144	0.069	0.052
L0H2	0.922	1.000	0.093	0.134	0.112	0.051
L1H0	0.027	0.093	1.000	0.812	0.952	0.451
L1H1	0.144	0.134	0.812	1.000	0.901	0.476
L1H3	0.069	0.112	0.952	0.901	1.000	0.468
L3H3	0.052	0.051	0.451	0.476	0.468	1.000

Two clusters are immediately visible. The Layer 0 heads L0H1 and L0H2 have cosine

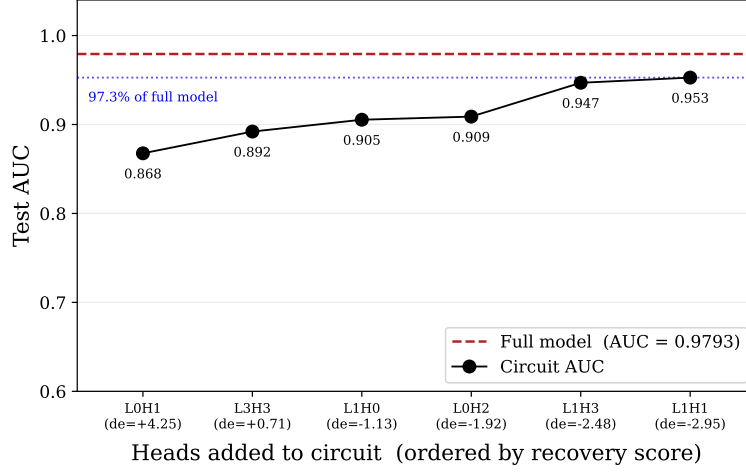


Figure 8: Partial minimality test. Test AUC of the partial circuit as heads are added one at a time in order of decreasing recovery score. Error bars show the 95% bootstrap confidence interval. The dashed red line shows the full model AUC (0.9793). The two largest single-step gains occur when the primary source L0H1 is added (step 1, +0.868 AUC) and when the relay head L1H3 is added (step 5, +3.8 percentage points), the latter only after the upstream context from L0H1, L3H3, L1H0, and L0H2 is in place.

similarity 0.922, indicating that their mean output representations are nearly parallel: this justifies the *secondary source* taxonomy assigned to L0H2, in contrast to the qualitatively different (and orthogonal-to-source) representations of the Layer 1 relay heads. The three Layer 1 relay heads form a tight cluster with pairwise cosine similarities 0.812–0.952, consistent with their playing redundant relay roles. The L3H3 readout has cosine similarity at most 0.476 to any other circuit head, consistent with performing a qualitatively distinct aggregation operation.

The pairwise zero-ablation analysis quantifies the redundancy. Table 7 reports, for all $\binom{6}{2} = 15$ pairs of circuit heads, the joint-ablation damage, the sum of the two individual-ablation damages, and their difference (the super-additivity). A negative super-additivity indicates that masking both heads together causes less damage than the sum of masking each individually, which is the expected signature of redundancy: when one head is masked, the other partially compensates by carrying a similar representation.

The pattern in Table 7 confirms and refines the representational similarity analysis. The two strongest redundancies ($S = -1.51$ for $\{L0H1, L0H2\}$ and $S = -1.50$ for both $\{L1H0, L1H3\}$ and $\{L1H1, L1H3\}$) are precisely the pairs identified as representational near-parallel by the cosine similarity in Table 6 (cosine similarities 0.92, 0.95, and 0.90 respectively). All five pairings involving L0H1 show negative super-additivity (S in $[-1.51, -0.31]$), reflecting that L0H1 is so dominant that the rest of the circuit partially compensates for its absence in any pairwise masking. By contrast, cross-cluster pairings that mix a Layer 0 source with an off-source-cluster Layer 1 relay or with the L3H3 readout are close to independent ($|S| \leq 0.18$), indicating that the Layer 0 and Layer 1 clusters carry largely non-overlapping computational roles, as already suggested by the representational orthogonality in Table 6 (cosine similarities below 0.15 between the two clusters). The readout head L3H3, in turn, is approximately independent of every relay and secondary-source head it is paired with (S in $[-0.13, +0.00]$),

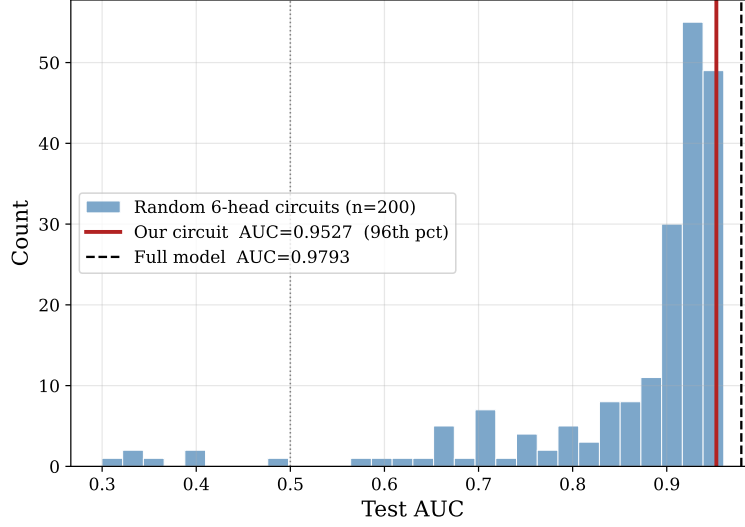


Figure 9: Random-baseline comparison. Test AUC distribution for 200 randomly sampled six-head circuits (blue histogram), compared to the ablation-identified circuit (red vertical line, $\text{AUC} = 0.9527$) and the full model (black dashed line, $\text{AUC} = 0.9793$). The identified circuit lies at the 96th percentile of the random distribution. The random distribution has substantial spread ($\sigma = 0.125$), with several subsets exceeding $\text{AUC} = 0.93$ and others falling below 0.65, indicating significant variation across random sparse subsets and a partly redundant learned representation.

consistent with its qualitatively distinct aggregation role.

6.8 Robustness across kinematic regimes

Table 8 compares the full model and circuit-only AUC in bins of jet multiplicity and jet mass on 10,000 test jets.

The circuit retains 90.6–99.4% of full-model performance across all six regimes. The largest degradation occurs for dense jets (92.6% retention) and low-mass jets (90.6%), both of which are kinematically challenging regimes in which the full model itself has a smaller advantage over classical methods. The high retention in sparse jets (99.4%) is consistent with the expectation that the circuit, which processes pairwise interactions through six attention heads, operates most effectively when the relevant pairs are not obscured by combinatorial background. The relatively uniform retention across the three jet-mass bins (90.6–92.4%) indicates that the circuit is not biased toward any particular mass range.

7 Physical Content of the Residual Stream

The circuit analysis of Section 6 identifies which attention heads carry the model’s discriminating computation but does not say what physical information those heads construct in the residual stream. In this section we address that question directly, by training linear probes that predict classical jet-substructure observables from the mean-pooled residual stream at each representation depth.

Table 7: Pairwise zero-ablation analysis for all $\binom{6}{2} = 15$ pairs of circuit heads. “Joint” is the drop in mean logit difference when both heads are masked simultaneously; “Sum” is the sum of the corresponding individual-ablation drops; the super-additivity S is their difference. Negative S indicates redundancy (joint masking damages less than the sum of individual damages); $S \approx 0$ indicates approximately independent contributions. The pair {L0H1, L0H2} is the most strongly redundant ($S = -1.51$), consistent with their high cosine similarity (Table 6); the pairs within the Layer 1 relay cluster show similarly large redundancy. Pairs that combine a Layer 0 head with a Layer 1 relay head (off-cluster pairings) are approximately independent.

Pair	Joint	Sum	Super-additivity S
<i>Within Layer 0 (source cluster):</i>			
{L0H1, L0H2}	0.732	2.244	-1.512
<i>Within Layer 1 (relay cluster):</i>			
{L1H0, L1H3}	0.738	2.236	-1.498
{L1H1, L1H3}	0.638	2.136	-1.498
{L1H0, L1H1}	0.638	1.375	-0.738
<i>L0H1 paired with each remaining circuit head:</i>			
{L0H1, L1H3}	2.570	3.011	-0.441
{L0H1, L3H3}	1.725	2.057	-0.332
{L0H1, L1H1}	1.841	2.150	-0.309
{L0H1, L1H0}	1.943	2.250	-0.307
<i>Cross-cluster (Layer 0 $\times$ Layer 1 off-cluster):</i>			
{L0H2, L1H0}	1.293	1.469	-0.177
{L0H2, L1H1}	1.328	1.369	-0.041
{L1H3, L0H2}	2.250	2.230	+0.020
<i>With readout head L3H3:</i>			
{L1H0, L3H3}	1.151	1.282	-0.131
{L0H2, L3H3}	1.227	1.276	-0.049
{L1H3, L3H3}	2.019	2.043	-0.024
{L1H1, L3H3}	1.184	1.182	+0.002

7.1 Linear probing of jet-level observables

We characterize the physical information encoded in the residual stream at each layer by training Ridge regression linear probes to predict jet-level observables from the mean-pooled particle representation. Each probe is trained on 7,000 jets and evaluated on 3,000 held-out jets; features are standardized before regression and the regularization strength is set to $\alpha = 1.0$.

Figure 10 reports the probe R^2 as a function of depth. The probe results reveal a clear hierarchy. Jet mass is encoded with $R^2 = 0.881$ at the embedding layer L0 and reaches $R^2 = 0.974$ by L1, remaining flat thereafter; this is consistent with jet mass being a near-additive function of the per-particle features. The leading- p_T fraction follows a similar pattern, reaching $R^2 = 0.858$ at L1.

The N -subjettiness ratios require more processing. τ_{21} rises from 0.332 at L0 to 0.713 at L2; τ_{32} rises more slowly, from 0.342 at L0 to 0.660 at L4. Neither τ_{32} nor τ_{21} saturates

Table 8: Full model and circuit-only AUC in three jet-multiplicity bins and three jet-mass bins, evaluated on 10,000 test jets. Retention is the ratio of the circuit AUC to the full model AUC in the same bin. The circuit retains at least 90% of the full model performance in every regime tested.

Regime	Full AUC	Circuit AUC	Retention
Sparse jets (few particles)	0.997	0.991	99.4%
Medium multiplicity	0.978	0.960	98.2%
Dense jets (many particles)	0.943	0.873	92.6%
Jet mass (low third)	0.948	0.858	90.6%
Jet mass (middle third)	0.950	0.870	91.6%
Jet mass (high third)	0.921	0.850	92.4%

within the particle attention stack at the linear-probe level, consistent with these observables requiring multi-particle reasoning that the attention mechanism progressively constructs.

The binary `is_top` probe reaches $R^2 = 0.868$ already at L0 and rises to $R^2 = 0.929$ at Cls0, with no further improvement at Cls1 (0.927). The flatness of the `is_top` trajectory across the particle attention layers, when read together with the basis-rotation analysis in Section 5.2, supports the same mechanistic picture: the class signal is fully present in the residual stream by L1 in some basis, and the class attention block aligns it with the head’s basis without introducing significant new information.

7.2 Energy correlator probes: 2-prong versus 3-prong encoding

The classical energy correlation function family contains observables designed for substructure of different prong multiplicities. The two-point correlator $C_1^{(\beta)}$ is most sensitive to single-prong versus diffuse radiation, the double ratio $C_2^{(\beta)}$ and the two-prong-optimal observable $D_2^{(\beta)}$, given in Eq. (10), target two-prong substructure such as a hadronic W or Z decay, and the next-order observables $C_3^{(\beta)}$ and $N_3^{(\beta)}$, as described in Appendix A.1 by Eq. 11 and Eq. 12 respectively, target three-prong substructure such as the full $t \rightarrow Wb \rightarrow q\bar{q}b$ decay topology [12]. In Eq. (12), ${}_v e_N^{(\beta)}$ denotes the generalized correlator that uses the v smallest pairwise angles among the $\binom{N}{2}$ pairs in each N -particle subset, normalized by the energy fractions $z_i = p_{T,i}/\text{ECF}(1, \beta)$. By construction $C_3^{(\beta)}$ and $N_3^{(\beta)}$ are the canonical IRC-safe observables for a 3-prong tagging task: they are sensitive to the full three-body kinematics of the top decay in a way different from the 2-prong observables.

We compute all the above observables for $\beta \in \{1, 2\}$ on 5,000 test jets, with $C_3^{(\beta)}$ and $N_3^{(\beta)}$ evaluated on the leading 50 and 40 particles per jet respectively for computational tractability ($\text{ECF}(4)$ and ${}_{2e_4}$ are $\mathcal{O}(n^4)$ in the constituent count). Linear probe R^2 values are reported in Table 9 for the full set of energy correlator and reference observables.

The probe values exhibit a clean and systematic ordering. Within the energy correlator family, the 2-prong observables are encoded more strongly than the 3-prong observables: the peak R^2 for $C_2^{(\beta=1)}$ is 0.945, compared to 0.804 for $N_3^{(\beta=1)}$ and 0.672 for $C_3^{(\beta=1)}$. Within the 3-prong group, the generalized correlator $N_3^{(\beta=1)}$ is encoded more strongly than $C_3^{(\beta=1)}$, consistent with N_3 being more robust to soft contamination because it uses only the two

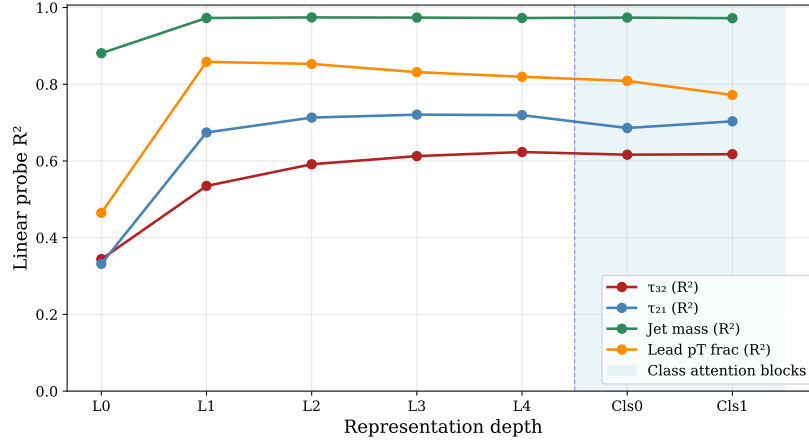


Figure 10: Linear probe R^2 for jet-level observables as a function of representation depth, including both particle attention layers (L0–L4) and class attention blocks (Cls0, Cls1). Jet mass and the leading- p_T fraction are encoded with high R^2 from the embedding layer; τ_{32} and τ_{21} improve more gradually through the particle attention stack. The binary `is_top` probe shows only modest improvement between L4 and the class attention blocks, consistent with the basis-rotation interpretation of Section 5.2.

smallest pairwise angles in each 4-particle subset. Both 3-prong observables are encoded more strongly than the corresponding N -subjettiness ratio τ_{32} (peak $R^2 = 0.581$).

The model encodes 2-prong structure more strongly than 3-prong structure. The most informative comparison is between the 3-prong observables C_3 and N_3 , which are the formally correct targets for a top tagger, and the 2-prong observables C_2 and D_2 , which target the substructure of a single hadronic two-body decay. Despite being trained on a 3-prong classification task, the model’s residual stream encodes 2-prong observables more strongly than 3-prong observables at every layer beyond L0. This is not a small effect: at the peak layer the gap is $|\Delta R^2| \approx 0.14$ between C_2 and N_3 , and $|\Delta R^2| \approx 0.17$ between D_2 and C_3 .

We interpret this phenomenon as the model implicitly factorising the top-tagging task into a more accessible sub-problem. In the kinematic regime studied here, $p_T \in [550, 650]$ GeV, the top decay products are collimated within an angular cone of $\Delta R \sim 2m_t/p_T \approx 0.5$, comfortably inside the $R = 0.8$ jet, and the visible substructure of the top jet contains a hard, heavy two-prong sub-system (the hadronic W decay, $W \rightarrow q\bar{q}$, with $m_W \approx 80$ GeV) recoiling against a b -quark sub-jet. A 2-prong tagger applied to such a jet identifies the W system through its heavy invariant mass; the full three-body topology, although in principle more discriminating, is more demanding on the relative angular arrangement of all three sub-jets and is more sensitive to soft contamination. The model has converged on the easier of the two available tagging strategies. The relay heads in the identified circuit (Section 6.2) attend selectively to high-invariant-mass particle pairs (Section 8.1) with positive class-discriminating power on the pairwise ECF(2) contribution (Section 8.3); these are precisely the kinematic features of the heaviest single pair in the jet, which in a top jet is dominated by the W -decay products. The 2-prong-over-3-prong encoding ordering is the global linear-probe signature of this local attentional bias.

Table 9: Linear probe R^2 for the energy correlator observables and classical reference observables at each representation depth. The peak R^2 across layers is given in the rightmost column, with the layer at which the peak is achieved indicated. The 3-prong observables $C_3^{(\beta=1)}$ and $N_3^{(\beta=1)}$, despite being the canonical targets for a 3-prong tagging task, are encoded less strongly than the corresponding 2-prong observables $C_2^{(\beta=1)}$ and $D_2^{(\beta=1)}$.

Observable	L0	L1	L2	L3	L4	Cls0	Cls1	Peak R^2	Peak layer
<i>2-prong energy correlators:</i>									
$C_1^{(\beta=1)}$	0.856	0.986	0.989	0.988	0.989	0.987	0.986	0.989	L2
$C_2^{(\beta=1)}$	0.670	0.915	0.930	0.941	0.945	0.936	0.935	0.945	L4
$D_2^{(\beta=1)}$	0.639	0.839	0.842	0.817	0.804	0.845	0.837	0.845	Cls0
<i>3-prong energy correlators:</i>									
$C_3^{(\beta=1)}$	0.303	0.648	0.672	0.662	0.623	0.639	0.594	0.672	L2
$N_3^{(\beta=1)}$	0.374	0.756	0.804	0.790	0.761	0.772	0.751	0.804	L2
<i>N-subjettiness reference:</i>									
τ_{21}	0.189	0.597	0.663	0.655	0.672	0.641	0.646	0.672	L4
τ_{32}	0.310	0.465	0.544	0.574	0.581	0.578	0.571	0.581	L4
Jet mass	0.862	0.969	0.972	0.971	0.971	0.971	0.969	0.972	L2

Energy correlator basis preferred over N -subjettiness. A complementary comparison is between the energy correlator basis and the N -subjettiness basis at fixed prong-count target. For 3-prong substructure, $N_3^{(\beta=1)}$ ($R^2 = 0.804$) is encoded substantially more strongly than τ_{32} ($R^2 = 0.581$); for 2-prong substructure, $C_2^{(\beta=1)}$ ($R^2 = 0.945$) is encoded substantially more strongly than τ_{21} ($R^2 = 0.672$). The energy correlator basis, which is constructed from pairwise (and higher) products of p_T and ΔR , is more closely aligned with the pairwise structure of the Particle Transformer’s interaction bias than the axis-based N -subjettiness construction. The model has discovered a representation that is closer to the energy correlator basis without either basis being explicitly encoded in the training signal.

Mass-residualization control. A potential concern is that the apparent encoding advantages above are driven by jet mass, which is itself encoded with $R^2 = 0.972$ already at L1 (Table 9). Of the observables discussed here, D_2 is the most strongly mass-correlated by construction: a coefficient of determination of 0.59 is obtained when D_2 is regressed on jet mass alone, compared to 0.26 for τ_{32} . To rule out mass as the driver of the D_2 -over- τ_{32} encoding gap, we residualize both observables against jet mass via a degree-3 polynomial regression fitted on the training subset and probe the residuals; the results are reported in Table 10. After residualization, the peak R^2 for $D_2^{(\beta=1)}$ is 0.807 and for τ_{32} is 0.531; the encoding advantage of D_2 over τ_{32} widens from 0.243 raw to 0.276 after the contribution from jet mass has been removed. The energy-correlator basis preference is therefore not a mass artefact.

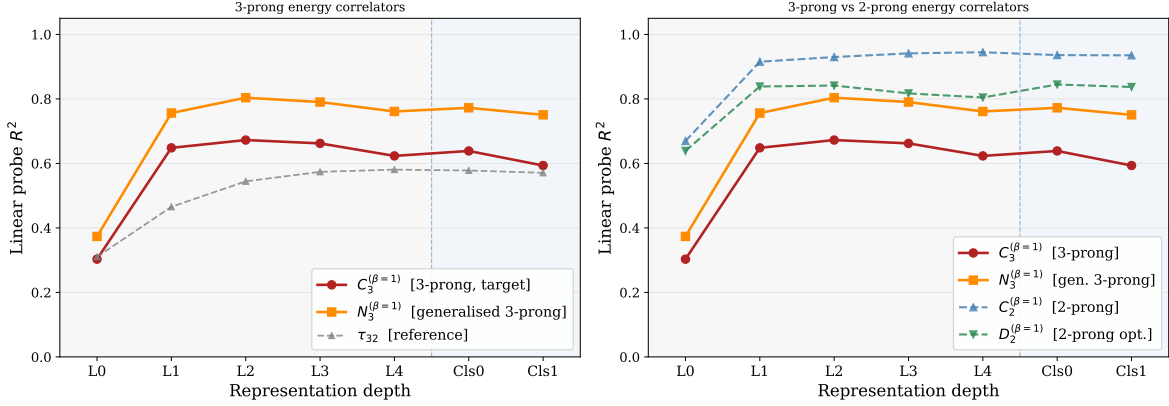


Figure 11: Linear probe R^2 for energy correlator observables as a function of representation depth. *Left*: the 3-prong observables $C_3^{(\beta=1)}$ (red) and $N_3^{(\beta=1)}$ (orange) against the N -subjettiness reference τ_{32} (grey, dashed). *Right*: the same 3-prong observables compared to the 2-prong observables $C_2^{(\beta=1)}$ (blue, dashed) and $D_2^{(\beta=1)}$ (green, dashed). The shaded region marks the class attention blocks. The 3-prong observables are encoded significantly more strongly than τ_{32} but significantly less strongly than the corresponding 2-prong observables, despite being the canonical targets for the 3-prong tagging task.

8 Interaction Features Driving Each Head

8.1 Correlation between attention and pairwise features

The attention bias $U_{ij}^{(h)}$ in Eq. (13) is computed by the pairwise MLP from the four pairwise features $\ln k_T$, $\ln z$, $\ln \Delta$, and $\ln m^2$. To understand which feature each head responds to, we measure the Pearson correlation between the attention weight $A_{ij}^{(l,h)}$ and each of the four features over all off-diagonal real particle pairs across 2,000 test jets. Table 11 reports the result for the four heads with the most substantial direct effects (excluding the readout head L3H3 and the Layer 1 head L1H1, whose ECF-attention correlations are discussed in Section 8.3).

L0H1 stands apart with negative correlations on $\ln k_T$ (-0.302), $\ln \Delta$ (-0.481), and $\ln m^2$ (-0.429): it preferentially attends to pairs with small angular separation, low transverse-momentum product, and low pairwise invariant mass. This is the kinematic regime of soft and collinear emissions, not of hard substructure. We return to the interpretation of this pattern in Section 9.1.

The other three heads (L0H2, L1H0, L1H3) show a common pattern of positive correlations with $\ln m^2$ ($+0.44$ to $+0.50$) and $\ln k_T$ ($+0.25$ to $+0.38$) and negative correlations with $\ln z$ (-0.15 to -0.31). These heads preferentially attend to particle pairs with large invariant mass, high k_T , and asymmetric momentum sharing, which are precisely the kinematic properties of decay products from a heavy particle. The hadronic decay products of the W boson in $t \rightarrow Wb \rightarrow q\bar{q}b$ provide the dominant source of large-invariant-mass, high- k_T pairs in a top jet, and the relay heads attention pattern is consistent with localizing on these pairs.

Table 10: Linear probe R^2 for $D_2^{(\beta=1)}$ and τ_{32} in their raw form and after residualization against jet mass. This test uses 7k train and 3k test jets, larger than for Table 9 mainly because of cheaper computation requirements. Mass alone explains a coefficient of determination of 0.59 for D_2 and 0.26 for τ_{32} ; both observables retain substantial mass-independent information. After residualization, the D_2 encoding remains well above the τ_{32} encoding at every layer, confirming that the model’s preferential encoding of D_2 is not a mass artefact.

Observable	L0	L1	L2	L3	L4	Cls0	Cls1	Peak R^2
$D_2^{(\beta=1)}$ raw	0.741	0.880	0.894	0.889	0.883	0.901	0.903	0.903
$D_2^{(\beta=1)}$ mass-residualized	0.368	0.767	0.796	0.790	0.785	0.800	0.807	0.807
τ_{32} raw	0.342	0.561	0.600	0.633	0.660	0.650	0.645	0.660
τ_{32} mass-residualized	0.141	0.397	0.454	0.496	0.531	0.518	0.515	0.531

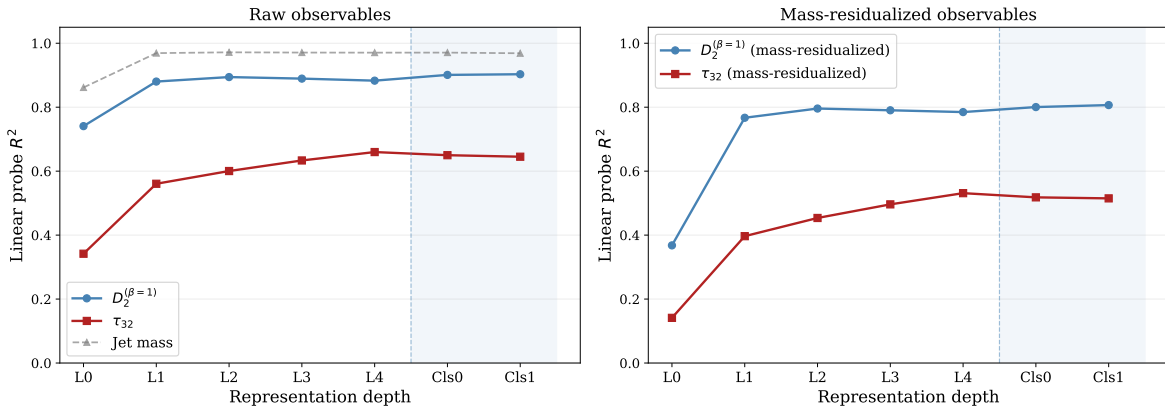


Figure 12: Linear probe R^2 for $D_2^{(\beta=1)}$ (blue) and τ_{32} (red) in raw form (left) and after residualization against jet mass (right). The dashed grey line on the left panel shows the jet mass probe R^2 for reference. The D_2 -over- τ_{32} encoding advantage is preserved, and indeed widens slightly, after the contribution from jet mass has been removed.

8.2 Causal feature ablation

The Pearson correlations of Section 8.1 are correlational. To establish causal dependence, we zero each pairwise feature individually in the input to the pairwise MLP, re-run the model, and measure the resulting change in the logit difference and in the attention pattern.

Figure 13 shows the effect of each single-feature ablation on the top-versus-QCD attention difference profile of four heads. Zeroing $\ln k_T$ produces the largest change in the mean logit difference across all single-feature ablations (+4.44 when zeroed, where the convention is that a positive change is a degradation), identifying $\ln k_T$ as the most causally important pairwise feature for the model’s overall performance. The relative profile changes are substantial for all four heads (relative L^2 change in the ΔR -resolved attention difference: 0.86–1.44), consistent with $\ln k_T$ being a primary driver of attention patterns across heads.

The ablation also reveals head-specific dependencies. Zeroing $\ln m^2$ produces a disproportionately large profile change in the relay-type panels relative to the source-type panel, consistent with the correlation pattern in Table 11 showing that the relay heads have stronger

Table 11: Pearson correlation between attention weights $A_{ij}^{(l,h)}$ and the four pairwise interaction features for four representative circuit heads. Correlations are pooled over all off-diagonal real particle pairs in 2,000 test jets. L0H1 stands apart with negative correlations on $\ln k_T$, $\ln \Delta$, and $\ln m^2$; the other three heads show positive correlations on $\ln k_T$ and $\ln m^2$ and negative correlations on $\ln z$.

Head	$\ln k_T$	$\ln z$	$\ln \Delta$	$\ln m^2$
L0H1	-0.302	+0.152	-0.481	-0.429
L0H2	+0.384	-0.152	+0.185	+0.504
L1H0	+0.250	-0.308	+0.108	+0.439
L1H3	+0.315	-0.201	+0.141	+0.454

$\ln m^2$ correlations than L0H1.

A counter-intuitive observation is that zeroing $\ln z$ produces a *positive* change in the mean logit difference (+2.69), meaning that the model performs slightly better when this feature is removed. A possible interpretation can be that $\ln z$, which measures the momentum sharing asymmetry of a pair, can drive the attention onto soft fragments with high z asymmetry but small absolute momentum scale, and removing this input partially suppresses a confounding signal. We do not interpret this as a sign that $\ln z$ should be removed from the architecture in practice; the effect is small compared to the $\ln k_T$ effect, and more importantly the $\ln z$ feature is a non-trivial component of the representation underlying the relay heads' positive direct effects on $\ln m^2$, which would also be affected by its removal in non-trivial ways.

8.3 ECF-attention correlation and class-discriminating attention

The interaction-feature correlations of Section 8.1 characterize each head's attention pattern at the level of single pairwise features ($\ln k_T$, $\ln z$, $\ln \Delta$, $\ln m^2$). A complementary characterization, more directly tied to the substructure observables of Section 7.2, asks whether the head attends selectively to particle pairs that participate in higher-order energy correlator structure, and whether this selective attention is class-dependent.

For each pair (i, j) in a jet we construct two scalar proxies that quantify the contribution of that pair to substructure of a given prong-count. The 2-prong proxy is the (normalized) pairwise contribution to $\text{ECF}(2, \beta)$,

$$\tilde{c}_{2;ij} = \frac{p_{T,i} p_{T,j} \Delta R_{ij}^\beta}{\text{ECF}(1)^2}, \quad (3)$$

which depends only on the kinematics of the pair (i, j) itself. The 3-prong proxy is the pair's marginal contribution to $\text{ECF}(3, \beta)$, obtained by summing the energy correlator weight over the choice of third particle k ,

$$\tilde{c}_{3;ij} = \frac{p_{T,i} p_{T,j} \Delta R_{ij}^\beta}{\text{ECF}(1)^3} \sum_{k \neq i,j} p_{T,k} \Delta R_{ik}^\beta \Delta R_{jk}^\beta. \quad (4)$$

By construction, $\tilde{c}_{3;ij}$ is large when the pair (i, j) sits at the apex of a 3-particle configuration with a third high- p_T particle at small angular separation from both i and j . In a top jet, the

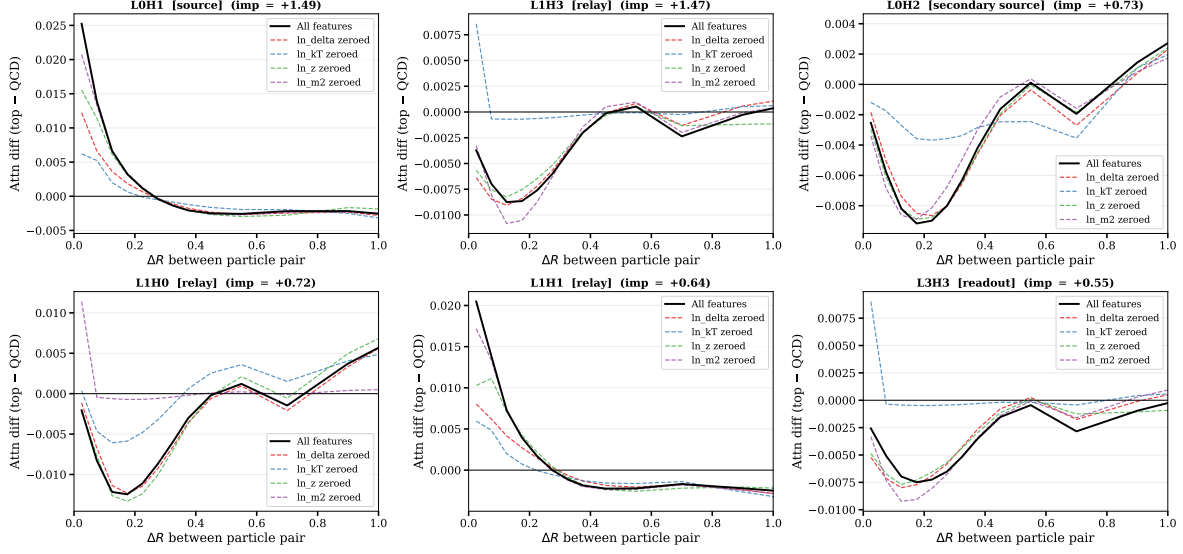


Figure 13: Causal feature ablation. For each of the four pairwise features (coloured dashed lines), the feature is zeroed in the input tensor before the pairwise MLP and the model is re-run. The plotted quantity is the difference in the head’s mean attention weight between top and QCD jets as a function of pairwise ΔR . A large deviation from the baseline (black) indicates that the head causally depends on that feature. Profiles are averaged over 10,000 test jets. The six panels correspond to six heads spanning a range of zero-ablation importances, the value of which is given in each panel title.

dominant such configurations are those involving the three hard sub-jets of the $t \rightarrow Wb \rightarrow q\bar{q}b$ decay; in a QCD jet, they are produced through ordinary parton-shower evolution. The two proxies $\tilde{c}_{2;ij}$ and $\tilde{c}_{3;ij}$ are not mutually exclusive: every pair has both, and the proxies factorize the contribution of pair (i, j) to the energy correlator structure of the jet by prong-count.

For each circuit head (l, h) and each prong-count $n \in \{2, 3\}$ we compute the Pearson correlation $r_n(l, h)$ between the head’s attention weight $A_{ij}^{(l, h)}$ and the corresponding proxy $\tilde{c}_{n;ij}$, separately for top jets and for QCD jets. The class-discriminating power of the head on the n -prong proxy is then defined as

$$\delta_n(l, h) \equiv r_n^{\text{top}}(l, h) - r_n^{\text{QCD}}(l, h). \quad (5)$$

A positive δ_n indicates that the head’s attention is more selective for n -prong-relevant pairs in top jets than in QCD jets; $\delta_n \approx 0$ indicates that the head’s attention is class-agnostic at the level of n -prong pairwise contributions. The pair (δ_2, δ_3) characterizes whether the head’s class-discriminating attention operates predominantly at the 2-particle or the 3-particle level.

Table 12 reports both proxies for the six circuit heads, computed over 2,000 test jets; Figure 14 displays the same data graphically.

The six circuit heads separate into two clear groups. The primary source L0H1 and the relay head L1H1 have δ_2 and δ_3 both close to zero (and slightly negative), indicating that their attention is class-agnostic at the level of pairwise correlator contributions. This is consistent with the interaction-feature analysis of Section 8.1: L0H1’s attention is biased toward soft and collinear pairs, which are not the pairs that contribute to either the 2-prong or the 3-prong proxy in a class-discriminating way.

Table 12: Pearson correlations between attention weights and the 2-prong and 3-prong pairwise proxies of Eqs. (3)–(4), computed separately on top and QCD jets over 2,000 test jets. The class-discriminating powers $\delta_n = r_n^{\text{top}} - r_n^{\text{QCD}}$ measure how selectively each head attends to n -prong pairwise contributions in top jets relative to QCD jets. For four of the six circuit heads (L0H2, L1H0, L1H3, L3H3), the 3-prong discriminating power δ_3 exceeds the 2-prong discriminating power δ_2 .

Head	r_2^{top}	r_2^{QCD}	δ_2	r_3^{top}	r_3^{QCD}	δ_3
L0H1	−0.092	−0.071	−0.021	−0.103	−0.095	−0.007
L0H2	+0.538	+0.423	+0.114	+0.493	+0.279	+0.214
L1H0	+0.373	+0.320	+0.054	+0.370	+0.228	+0.141
L1H1	−0.101	−0.081	−0.020	−0.114	−0.103	−0.011
L1H3	+0.511	+0.498	+0.014	+0.470	+0.337	+0.134
L3H3	+0.552	+0.475	+0.078	+0.521	+0.335	+0.186

The remaining four heads, the secondary source L0H2, the relay heads L1H0 and L1H3, and the readout L3H3, all show positive class-discriminating power on both proxies, with the same systematic ordering: $\delta_3 > \delta_2$ in all four cases, with δ_3 exceeding δ_2 by factors of 1.9 (L0H2), 2.6 (L1H0), 9.6 (L1H3), and 2.4 (L3H3). The relay head L1H3 is particularly striking: it is class-agnostic on the 2-prong proxy ($\delta_2 = 0.014$) but has a sizeable class-discriminating power on the 3-prong proxy ($\delta_3 = 0.134$), indicating that its class-selective attention operates almost entirely at the 3-particle level.

This $\delta_3 > \delta_2$ hierarchy is a non-trivial finding when read alongside the linear probe analysis of Section 7.2, which showed that the residual stream encodes 2-prong observables more strongly than 3-prong observables (C_2 at $R^2 = 0.945$ versus N_3 at $R^2 = 0.804$, and D_2 at $R^2 = 0.845$ versus C_3 at $R^2 = 0.672$). The two findings appear in tension at first reading: how can the heads’ attention be more 3-prong-selective than 2-prong-selective if the resulting residual stream is more 2-prong-encoded than 3-prong-encoded?

The reconciliation is that the 3-prong selectivity lives in the attention computation, while the residual stream receives the per-particle write-out. The pairwise-bias mechanism in the Particle Transformer (Section A.2) gives every attention head access to all pairwise features simultaneously and allows the head to attend selectively to pairs (i, j) in a way that depends on the presence of a third particle k (i.e., on a 3-particle configuration), because the bias $U_{ij}^{(h)}$ is constructed from features that implicitly encode this dependence. The output that each head writes to the residual stream, however, is a per-particle vector, and the linear probes of Section 7.2 read these representations in mean-pooled (per-jet) form. The 3-prong selection mechanism is therefore consumed in the attention computation but is not preserved as an explicit 3-prong representation in the residual stream; what is preserved is a 2-prong-dominant readout whose amplitude is modulated by 3-prong-aware attention. The 2-prong encoding seen by the linear probes and the 3-prong attention pattern seen by the class-discriminating-power analysis are therefore two sides of the same computation.

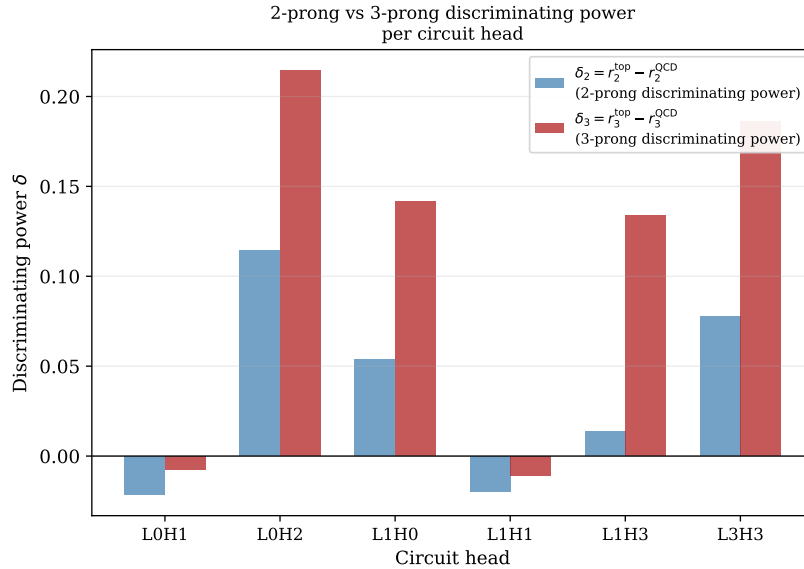


Figure 14: Class-discriminating powers δ_2 (blue) and δ_3 (red) for the six circuit heads, computed from the Pearson correlations between the head’s attention weights and the pairwise proxies of Eqs. (3)–(4) on top and QCD jets over 2,000 test jets. For L0H2, L1H0, L1H3, and L3H3, the 3-prong discriminating power exceeds the 2-prong discriminating power; for L0H1 and L1H1, both are negligible.

9 Discussion

In this section we take a step ahead to understand the inter-connection between all the results that we systematically obtained in the previous sections.

9.1 The circuit as a physical algorithm

Combining the evidence from the preceding sections, the Particle Transformer can be described as implementing top-quark tagging through a two-stage algorithm with the following structure.

Stage 1: feature extraction in the residual stream. The four particle attention layers act jointly to construct a near-complete classification representation in the residual stream by L1 (per-layer logistic probe AUC 0.971 at L1, rising marginally to 0.977 by L4; Section 5.2). This representation encodes two-prong energy correlator structure in particular: the genuine mass-residualized $D_2^{(\beta=1)}$ encoding rises from $R^2 = 0.368$ at L0 to $R^2 = 0.767$ at L1 (Section 7.2). Within this representation, the identified six-head circuit handles the bulk of the discriminating computation. Its components have qualitatively distinct roles:

- L0H1 (primary source) attends preferentially to soft and collinear particle pairs (negative $\ln k_T$, $\ln \Delta$, $\ln m^2$ correlations; Section 8.1) and has a class-agnostic relationship to ECF(2) at the level of pairwise contributions (Section 8.3). Despite this, it has the largest direct effect (+4.25) and is by itself sufficient to recover 88.6% of the full model AUC. We interpret L0H1 as performing a *soft-radiation contextualization* step: by summarizing the soft and collinear emission pattern in a class-agnostic way, it provides

the downstream relay heads with a normalization against which the hard substructure can be identified. The strong path effects from L0H1 to all three Layer 1 relay heads (1.19–1.45; Section 6.3) are consistent with this role.

- L0H2 (secondary source) is nearly parallel to L0H1 in representational space (cosine similarity 0.92; Section 6.7) but, in contrast to L0H1, has positive correlations with $\ln k_T$, $\ln m^2$ and a positive class-discriminating ECF(2) attention pattern ($\delta = +0.112$; Section 8.3). It plays the role of a class-discriminating complement to L0H1 in Layer 0, contributing weaker but still significant path effects to the same downstream targets (0.85–0.90).
- L1H0, L1H1, L1H3 (relay heads) all attend preferentially to high-invariant-mass, high- k_T , asymmetric- z pairs (Section 8.1) and form a tight representational cluster (cosine similarities 0.81–0.95). Their joint role is to localize the hard two-prong substructure characteristic of the W -boson decay $W \rightarrow q\bar{q}$. Their negative direct effects confirm that their clean-run activations are conditional on the upstream signal from the source heads; their large pairwise super-additivities confirm that they are partly redundant.
- L3H3 (readout) has low representational similarity to all other circuit heads (≤ 0.48), a positive direct effect of $+0.71$, and a positive class-discriminating ECF(2) attention ($\delta = +0.076$). It performs a final aggregation operation that combines the relayed signal into the format consumed by the class attention block.

Stage 2: basis alignment by the class attention block. The class attention block Cls0 produces the dramatic logit-lens jump from 0.111 to 0.973 in a single step (Section 5.1). The per-layer probe analysis resolves this jump as a basis rotation (Section 5.2): the linearly accessible class information is essentially complete by L1 in the particle attention basis, and Cls0 reorients this signal into the basis on which the final classification head was trained. The second class attention block Cls1 adds essentially no information ($+0.006$ logit-lens AUC), as expected for a basis-aligned representation.

This two-stage picture is conceptually compact and resembles the structure of circuits identified in language models in the IOI literature [16], in which an explicit name-mover computation is preceded by upstream backup and inhibition heads. The present work establishes that the source-relay-readout architectural pattern arises in a physics classifier trained on a physically motivated task without language-model-style structure being built into the data, and is therefore unlikely to be a feature specific to natural language.

9.2 Connection to classical jet substructure observables

The probe analysis of Section 7.2 shows that the model’s residual stream encodes $D_2^{(\beta=1)}$ to peak $R^2 = 0.903$ in raw form and 0.807 after residualization against jet mass, while τ_{32} is encoded to 0.660 raw and 0.531 residualized. The gap between D_2 and τ_{32} is not a consequence of the encoding of jet mass and persists, indeed widens, after mass is removed.

A natural interpretation is that the model has implicitly discovered a two-prong tagger as the dominant component of its top tagger. Within a top jet, the dominant pairs that contribute to large-invariant-mass configurations are the daughters of the hadronic W decay, which has $m_W \approx 80$ GeV. The model’s relay heads attend selectively to exactly these pairs (positive $\ln m^2$ correlations and positive ECF(2) class-discriminating power on L0H2, L1H0,

L3H3; Sections 8.1 and 8.3). The two-prong-optimal observable D_2 , which is constructed from ECF(2) and ECF(3) to be optimal under IRC-safe conditions for two-prong versus one-prong discrimination [45], matches this internal structure more closely than τ_{32} , which is constructed instead around the residual radiation about three-prong axes. The model has organized its computation around the most identifiable substructure, i.e. the two-prong W decay, rather than the full three-body topology of the top decay.

This is, to our knowledge, the first explicit demonstration of using causal circuit analysis for analysing jet-tagger interpretability.

9.3 Limitations

The analysis presented here has several limitations.

First, the model studied is a small Particle Transformer (4 particle attention layers, 4 heads per layer, 1.3M parameters) trained on the standard top-tagging benchmark. Larger and more capable Particle Transformers, trained at higher AUC (0.985 and above), may have richer circuit structures that are not captured by a six-head analysis. The methods developed here scale directly, as in studies pertaining natural language, but the specific six-head identification should not be assumed universal.

Second, the random-baseline comparison uses 200 randomly sampled six-head subsets out of $\binom{16}{6} = 8008$ possible subsets. This sampling is sufficient to place the identified circuit at the 96th percentile of the random distribution with empirical standard deviation $\sigma = 0.125$. However, a complete evaluation will provide the final word on this aspect.

Third, the linear probes used throughout the residual-stream analysis measure linear accessibility of information, not the full information content. Observables that are encoded non-linearly would be detected at lower R^2 than they truly are, and the values reported here should be read as lower bounds on the true encoding fidelity.

Fourth, the path-patching analysis uses two on-manifold corruption strategies in within-batch replacement, jet permutation as discussed in Section 6.6, and we have documented that off-manifold (Gaussian) corruption is structurally incompatible with the standard recovery-score formulation on this dataset. The qualitative sign pattern is robust within the on-manifold class; quantitative direct-effect magnitudes vary by up to a factor of two between strategies and should be interpreted with this caveat.

Fifth, although we have verified the cross-seed stability of the zero-ablation importance ranking across five training seeds, the detailed path-patching, probing, and feature-attribution analyses use the seed-0 model. Extending the path-patching analysis to multiple seeds would test whether the source-relay-readout sign pattern (rather than the specific head identities) is universal. We expect this to be the case based on the cross-seed stability of zero-ablation importance, but we leave a full multi-seed circuit verification to future work.

10 Conclusion

We have performed a complete mechanistic interpretability analysis of a Particle Transformer trained for top quark tagging on the standard benchmark dataset. Combining zero ablation across five training seeds, path patching with two complementary on-manifold corruption strategies, bootstrap-resolved path-effect estimates, linear probing of the residual stream, and

per-layer trained probes that resolve basis-rotation effects in the standard logit lens, we have arrived at the following picture of the model’s internal computation.

A sparse six-head circuit recovers 97.3% of the full model AUC and lies at the 96-th percentile of 200 randomly sampled six-head subsets. The circuit has a source-relay-readout structure: the primary source L0H1 carries nearly all the causal discriminating information (direct effect +4.25); the secondary source L0H2 contributes a weaker, complementary signal; three Layer 1 relay heads localize high-invariant-mass particle pairs and require the upstream context to function correctly; and the single Layer 3 readout head L3H3 aggregates the relayed signal. The sign pattern of the direct effects is robust under two on-manifold corruption strategies, and we have documented and explained a structural incompatibility between off-manifold (Gaussian) corruption and the standard recovery-score formulation on this kinematically narrow dataset.

The residual stream encodes the two-prong-optimal energy correlator $D_2^{(\beta=1)}$ to peak coefficient of determination $R^2 = 0.90$ in raw form and 0.81 after residualization against jet mass, both substantially above the corresponding values for N -subjettiness τ_{32} ($R^2 = 0.66$ and 0.53 respectively). The model has implicitly organized its computation around the two-prong substructure of the hadronic W decay, in agreement with the relay heads’ attention selectively to high-invariant-mass particle pairs.

Resolving the standard logit-lens analysis with per-layer trained logistic probes, we have shown that the apparent commitment of the model to a classification decision in the first class attention block is in fact closer to a basis rotation: the linearly accessible class signal saturates to $\text{AUC} \approx 0.97$ already in the first particle attention layer, and the class attention block aligns this signal with the basis on which the trained classifier head operates.

These results demonstrate that mechanistic interpretability methods developed in the context of natural language models transfer, with care, to jet physics classifiers. They reveal that a Particle Transformer trained on raw four-momenta can rediscover a physically meaningful energy-correlator structure through gradient descent alone, and they suggest that the source-relay-readout architecture observed here may be characteristic of a broader class of physics-domain Transformers. Future work should apply these methods to larger Particle Transformer variants and to other classification tasks, test the universality of the source-relay-readout architecture, and explore whether the basis-rotation interpretation of the class attention block generalizes beyond top tagging.

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A Appendix

A.1 Jet substructure for top quark tagging

A boosted top quark with transverse momentum $p_T \in [550, 650]$ GeV decays to a jet whose constituents encode the kinematics of the three-body final state $q\bar{q}b$. Two classical families of observables are particularly relevant to the present work.

N-subjettiness. The N -subjettiness observable [9, 10] is defined as

$$\tau_N = \frac{1}{d_0} \sum_k p_{T,k} \min_j \Delta R(k, \hat{n}_j)^\beta, \quad d_0 = \sum_k p_{T,k} R_0^\beta, \quad (6)$$

where the sum runs over jet constituents, the minimum is taken over N candidate subjet axes $\{\hat{n}_j\}$, R_0 is the jet radius, and β is an angular weighting exponent. The ratios $\tau_{21} \equiv \tau_2/\tau_1$ and $\tau_{32} \equiv \tau_3/\tau_2$ are sensitive to the presence of two-prong and three-prong substructure, respectively, and τ_{32} is the canonical N -subjettiness variable for top tagging.

Energy correlation functions. The (non-normalized) energy correlation functions [11] of order n and angular weight β are defined by

$$\text{ECF}(1, \beta) = \sum_i p_{T,i}, \quad (7)$$

$$\text{ECF}(2, \beta) = \sum_{i < j} p_{T,i} p_{T,j} \Delta R_{ij}^\beta, \quad (8)$$

$$\text{ECF}(3, \beta) = \sum_{i < j < k} p_{T,i} p_{T,j} p_{T,k} (\Delta R_{ij} \Delta R_{ik} \Delta R_{jk})^\beta. \quad (9)$$

The double ratios

$$C_2^{(\beta)} = \frac{\text{ECF}(3, \beta) \text{ECF}(1, \beta)}{\text{ECF}(2, \beta)^2}, \quad D_2^{(\beta)} = \frac{\text{ECF}(3, \beta) \text{ECF}(1, \beta)^3}{\text{ECF}(2, \beta)^3}, \quad (10)$$

have been shown to be optimal observables for two-prong discrimination under infrared- and collinear-safe (IRC) conditions [45, 46].

The next-order observables

$$C_3^{(\beta)} = \frac{\text{ECF}(4, \beta) \text{ECF}(2, \beta)}{\text{ECF}(3, \beta)^2} \quad (11)$$

and the generalized correlator

$$N_3^{(\beta)} = \frac{2e_4^{(\beta)}}{(e_3^{(\beta)})^2} \quad (12)$$

target three-prong substructure such as the full $t \rightarrow Wb \rightarrow q\bar{q}b$ decay topology [12].

We compute these observables in a pure NumPy [47] implementation using constituent four-momenta and conventional definition as in [11].

A.2 The Particle Transformer

The Particle Transformer [8] processes a jet of P constituent particles (zero-padded to a fixed maximum) using a stack of L *particle attention* blocks followed by two *class attention* blocks. The per-particle features are projected to a d -dimensional embedding by a multi-layer perceptron, yielding an initial residual stream $x^0 \in \mathbb{R}^{P \times d}$. The pairwise features are encoded by a stack of pointwise 1D convolutions (i.e., a position-wise MLP applied independently to each particle pair (i, j)) with GELU nonlinearities and batch normalization between layers, producing a shared pairwise embedding tensor $U \in \mathbb{R}^{P \times P \times d'}$, where d' is chosen equal to the number of attention heads N_{heads} . The same U is used as a bias by every particle attention block.

For head h in particle attention block l , the standard scaled dot-product attention is augmented by the h -th channel of U as a pre-softmax additive bias:

$$A_{ij}^{(l,h)} = \text{softmax}_j \left[\frac{Q_i^{(l,h)} K_j^{(l,h)\top}}{\sqrt{d_h}} + U_{ij}^{(h)} \right], \quad (13)$$

where $d_h = d/N_{\text{heads}}$ is the per-head embedding dimension and $Q^{(l,h)}, K^{(l,h)}$ are linear projections of the layer- l residual stream. The pairwise features feeding U are the four quantities

$$\ln k_{T,ij}, \quad \ln z_{ij}, \quad \ln \Delta_{ij}, \quad \ln m_{ij}^2, \quad (14)$$

with

$$k_{T,ij} = \min(p_{T,i}, p_{T,j}) \Delta R_{ij}, \quad (15)$$

$$z_{ij} = \frac{\min(p_{T,i}, p_{T,j})}{p_{T,i} + p_{T,j}}, \quad (16)$$

$$\Delta_{ij} = \Delta R_{ij}, \quad (17)$$

$$m_{ij}^2 = (p_i + p_j)^2, \quad (18)$$

where the invariant mass m_{ij}^2 is Lorentz-invariant in the full sense and the remaining three are invariant under longitudinal boosts. The pairwise bias allows each attention head to express a learned attention pattern that is informed by physical pairwise kinematics beyond what is available in the per-particle features alone.

Each particle attention block follows the NormFormer layout [44]: layer normalization both before and after the multi-head attention, layer normalization before each linear layer of the feed-forward sub-block, and residual connections around each sub-block. The output of each attention head is multiplied by a learned scalar gain $c_{l,h}$ before being summed into the residual stream; this is the head scaling that we exploit for zero ablation in Section 4.4.

The class attention blocks [48] aggregate the particle representations into a single classification vector via a learnable token $x_{\text{class}} \in \mathbb{R}^d$. In each class attention block, the multi-head attention takes its query from x_{class} alone, while the keys and values are computed from the concatenation $[x_{\text{class}}, x^L]$ of the class token with the output of the final particle attention block. The two class attention blocks act sequentially on x_{class} ; the final class token is passed through a layer normalization and a linear classification head producing two output logits for the top-versus-QCD classification.

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